

## **Pharmacology of Antifungal Agents**

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Following this session, students should be able to:

1. **PK:** Apply the pharmacokinetic profiles of the various systemic antifungal agents to their **clinical relevance in optimizing therapy** for the individual patient when given a case vignette.
2. **Mechanisms:** Compare the mechanisms of **action** and mechanisms of **resistance** of the various classes of antifungal agents using specific examples.
3. **Therapeutic uses:** Correlate the therapeutic uses with the antifungal spectrums and resistance patterns using specific examples.
4. **Adverse effects:** Relate the common and potentially serious adverse effects and precautions of the various antifungal agents to the management of antifungal therapy when given a case vignette.
5. **Drug interactions:** Explain the mechanisms and potential impact of drug interactions involving specific antifungal agents on the management of antifungal therapy in the individual patient using specific examples.

# Preparation Materials

## Required

- **Practice Questions**

Optional materials:

- Video lecture | Dr. Goldstein's Notes Handout | ScholarRx Bricks | Guided Reading Questions
- Textbooks and Examination Review Books (please see next slide)

## SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them. (ScholarRx Bricks & Practice Questions are required.)*
- *Working through the GUIDED READING QUESTIONS with pen/pencil and paper may help you recall what you have learned.*

*Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.*

- *Practice questions (not graded): Simple Recall and Case Vignettes*

### **Scholar Rx Brick:**

Name of Discipline/ Organ System: General Microbiology > Antimicrobial Agents > Antifungal Drugs

Title of Brick(s): Drugs to Treat Fungal Infections

Link(s): <https://exchange.scholarrx.com/brick/drugs-to-treat-fungal-infections>

Access Medicine Katzung's Basic & Clinical Pharmacology 16e, 2024: Chapter 48: Antifungal Agents

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382&sectionid=281755133>

Access Medicine Katzung's Pharmacology: Examination & Board Review 14e, 2024: Chapter 48: Antifungal Agents

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3461&sectionid=285598727>

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2022; Chapter 33: Antifungal Drugs

<https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329130&bookid=3222>

## Key points

- Fungi are eukaryotic, with rigid cell walls composed mainly of fibrillar polysaccharides, largely  $\beta$ -1,3-glucan and chitin, that bind covalently to each other and to proteins. The fungal cell membrane contains ergosterol rather than the cholesterol, as in mammalian cells.
- Fungal diseases are called mycoses and are often chronic infections. They may involve only the skin extending into the epidermis or may cause subcutaneous or systemic infections.
- Clinical response to antifungal treatment can be lengthy because fungi are complex slow-growing organisms with thick walls often embedded deep in tissues.
- Patients with chronic immune suppression due to organ transplantation, cancer chemotherapy, or HIV infection are susceptible to opportunistic fungal infections. Morbidity is high in immunosuppressed patients.
- Targets of antifungal drugs include components of the cell wall, ergosterol in the membrane, and nuclear functions.
- Drugs for the treatment of systemic mycoses include the polyene antifungals, azole antifungals, and echinocandins.
- Many azole antifungals and other structurally diverse drugs are used topically for superficial fungal infections.

### **Polyenes: Amphotericin B and Nystatin:** Amphipathic amphoteric molecules

Bind ergosterol → form pores in the membranes → leakage

Broad spectrum, Resistance is rare.

- Amphotericin B: I.V. | life-threatening infections | nephrotoxicity, electrolyte wasting, infusion rxns
- Lipid / liposomal formulations reduce toxicity.
- Nystatin: topical only due to severe toxicity, including nephrotoxicity, when given IV

**Flucytosine → 5-FU:** Inhibits fungal DNA and RNA synthesis → leads to impairment of protein synthesis and fungal cell proliferation

- Resistance: ↓ uptake by cytosine permease, ↓ conversion to active nucleoside analog by cytosine deaminase, ↓ conversion to toxic nucleotides, upregulated pyrimidine synthesis
- Narrow spectrum | oral only, high concentrations in CNS | always used in combination with another antifungal | bone marrow toxicity

**Azoles** (ketoconazole, itraconazole, fluconazole, voriconazole, isavuconazole, posaconazole)

- Inhibit lanosterol 14- $\alpha$ -demethylase → ↓ ergosterol synthesis in cell membrane → reduces membrane permeability, structural integrity, and impairs function
- Resistance: ERG11 mutations, efflux pumps
- Broad spectrum (in general) | topical treatment of superficial infection, oral/IV systemic infections
- ↑ serum liver enzymes/hepatotoxicity, QT prolongation, CYP interactions



## Key points: Targets and Drug Classes

**Echinocandins** (caspofungin, micafungin, anadulafungin): Inhibit  $\beta$ -1,3-D-glucan  $\rightarrow$  weakened cell wall  $\rightarrow$  osmotic influx of water  $\rightarrow$  cell swells  $\rightarrow$  cell ruptures (lysis)

- Narrow spectrum: *Candida* spp, *Aspergillus* | IV, poor CNS penetration | low toxicity, few DDIs

**Terbinafine**: Inhibits squalene epoxidase (an early step in ergosterol synthesis)  $\rightarrow$  reduces membrane permeability, structural integrity, and impairs function

- Resistance is rising: mutations in squalene epoxidase gene, efflux pumps
- Topical use for tinea, cutaneous candidiasis | systemically for fungal nail infections (onychomycosis)
- GI discomfort, headache, hepatotoxicity, rash | CYP2D6 inhibitor, moderate potential for drug interactions

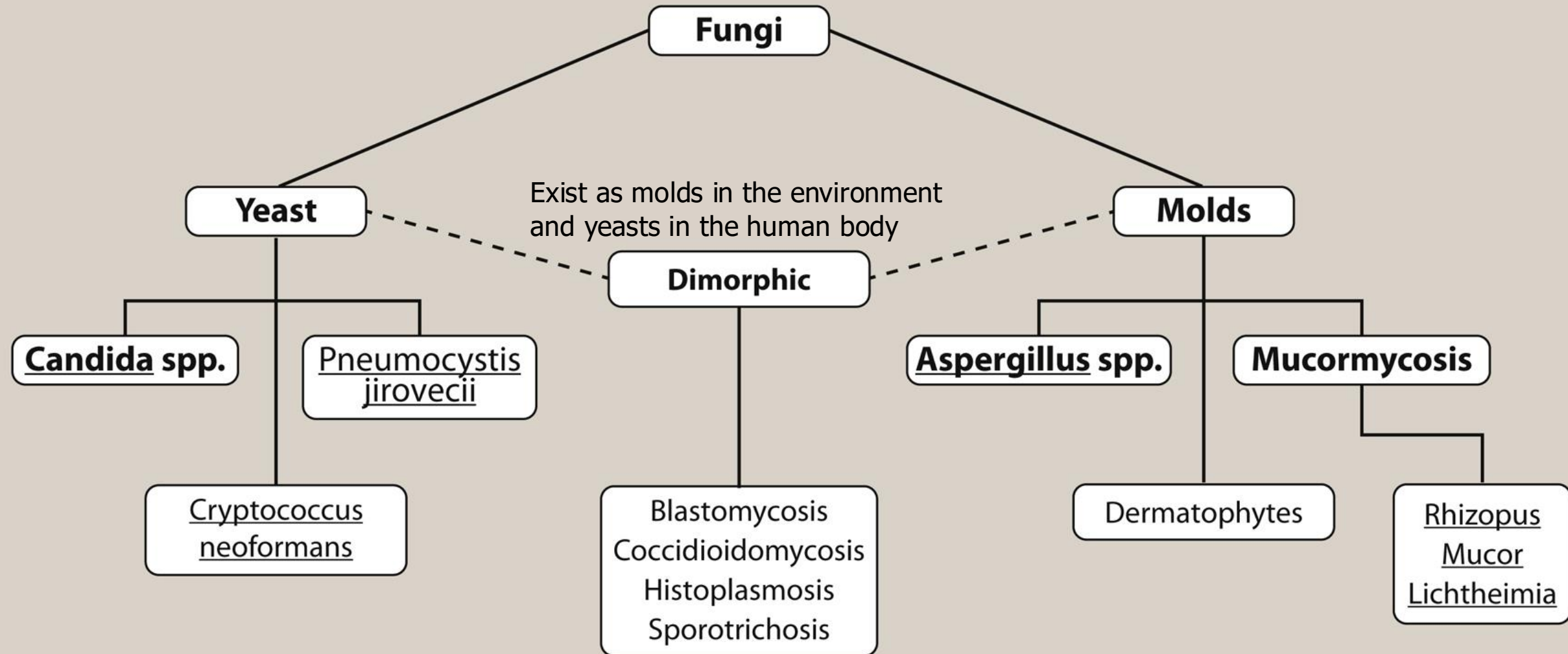
**Griseofulvin**: Binds microtubules in keratin precursor cells  $\rightarrow$  disrupts the mitotic spindle  $\rightarrow$  inhibits fungal cell mitosis

- Resistance: altered microtubule structure, decreased uptake of drug
- Narrow spectrum: tinea infections only (skin, hair, nails)
- Oral | | long duration of treatment | GI upset, headache,  $\uparrow$  serum liver enzymes | contraindicated in pregnancy



From: **33 Antifungal Drugs**

Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023



**Legend:** Common pathogenic organisms of Kingdom Fungi.

Clinical response to antifungal treatment can be protracted because fungi are complex slow-growing organisms with thick walls often embedded in tissues.

### Opportunistic fungal pathogens

- *Candida* spp.
  - *Aspergillus* spp.
  - *Cryptococcus neoformans*
  - *Pneumocystis jirovecii*
  - Mucorales order
    - *Rhizopus*
    - *Mucor*
    - *Rhizomucor*
    - and other species
- The most common causes of human infections in this group

### Endemic fungal pathogens

- *Blastomyces*: Midwest, south-central, southeastern regions
- *Coccidioides*: Southwestern US
- *Histoplasma*: Ohio and Mississippi River valleys
- *Paracoccidioides*: South America
- *Sporothrix*: worldwide
- others

- ❖ Opportunistic fungi are usually residents of the human microbiota or saprophytes in the environment. *Candida* and *Aspergillus* are the most common infections. Immunodeficient patients are at highest risk of opportunistic infection.
- ❖ Endemic fungi are not part of the normal human microbiome. Infection can result from inhalation of spores. (*Sporothrix* is through skin contact typically.) People who live in or travel to geographic areas where certain fungi are endemic are at risk for these infections. Weakened immune system or chronic lung disease increases the risk of endemic infection.
- ❖ The incidence of fungal infections is rising. Morbidity in immunocompromised patients is high.

**Ergosterol** is the primary fungal cell membrane sterol.  
**Cholesterol** is the primary human cell membrane sterol.

### Drug targets

Cell membrane: Ergosterol

- Amphotericin B (polyenes)
- Azoles
- Terbinafine

Cell-wall:  $\beta$ -glucans

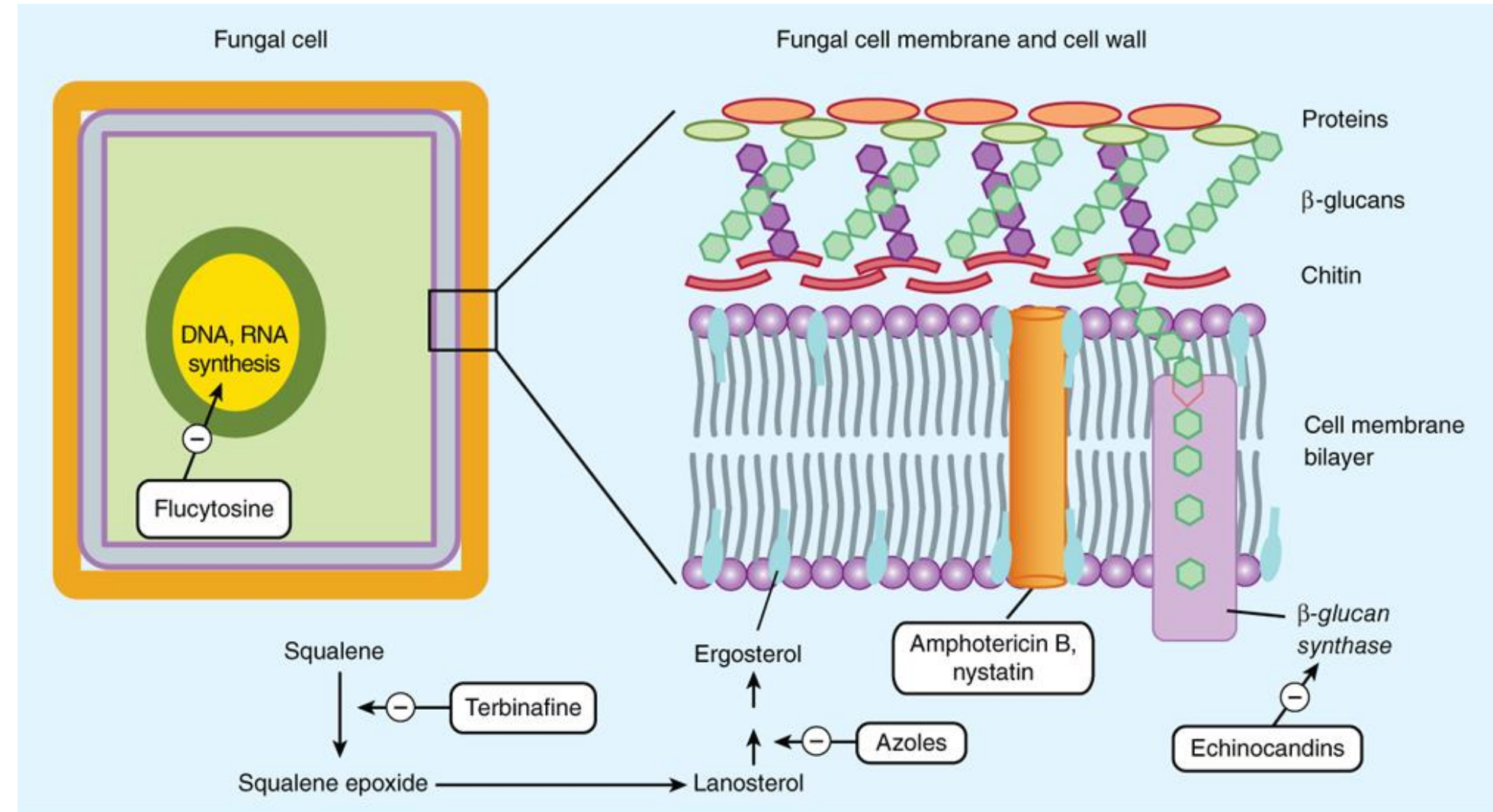
- Echinocandins

Nucleic acid synthesis

- Flucytosine

Microtubules

- Griseofulvin



B. G. Katzung, M. Kruidering-Hall, R. L. Tuan, T. W. Vanderah, A. J. Trevor  
*Katzung & Trevor's Pharmacology: Examination & Board Review, 13e*  
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Targets of antifungal drugs. Except for flucytosine and griseofulvin (not shown), all currently available antifungals target the fungal cell membrane or cell wall.

## Mechanisms of Resistance to Antifungal Agents

- Usually develops slowly
- Fungal cells do not transfer genes among or between species, and they do not express enzymes that inactivate antifungal agents.

The clinically important mechanisms of antifungal drug resistance are:

1. **Target protein alterations** due to genetic mutations that reduce binding affinity
2. **Expression of efflux pumps** that actively transport drug out of the fungal cell

Mechanisms vary according to the subclass of antifungal drugs.

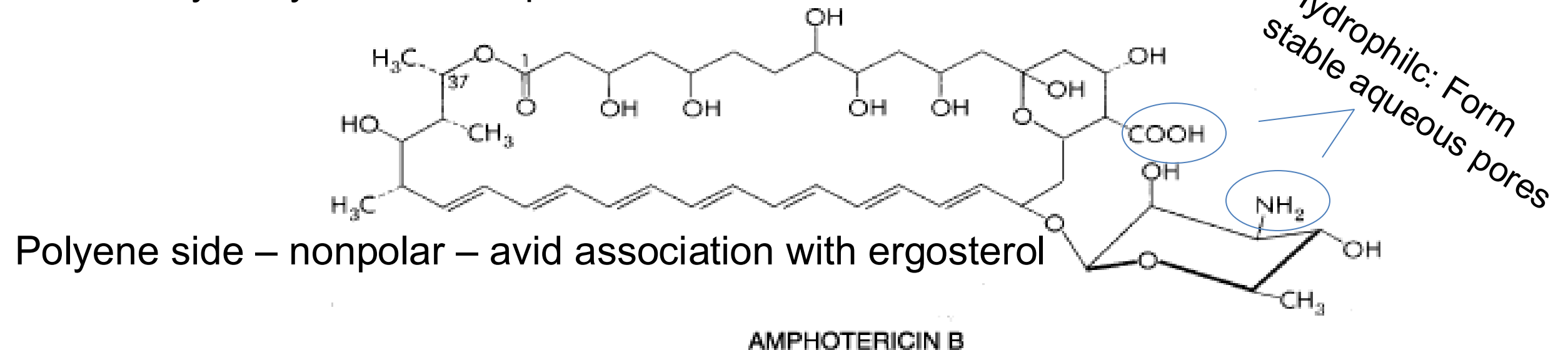
- *Candida albicans* resistance mechanisms are the most studied.
- Less is known about *Aspergillus* and *C. neoformans* resistance mechanisms.
- Almost no information on resistance mechanisms of other pathogenic fungi is available.

# \*Amphotericin B

an amphipathic, amphoteric polyene macrolide obtained from the bacterium *Streptomyces nodosus*

Broad spectrum, fungicidal

Hydroxyl-rich side – polar – associates with water molecules



- polyene: a hydrocarbon with several carbon–carbon double bonds, especially one having a chain of alternating single and double bonds
- amphipathic: the molecule has both hydrophobic and hydrophilic regions
- amphoteric: the molecule can act as either an acid or base



# Pharmacokinetics of Amphotericin B Formulations

**TAKEAWAYS:** IV only, highly protein bound, high concentration in tissues, poor penetration into CNS, slow elimination, cleared primarily by kidneys, long terminal half-life

Lipid / liposomal formulations decrease accumulation in kidney tissue → lower nephrotoxicity which allows higher dosing and improved tolerability

|           | Conventional AMB deoxycholate                                                                                                                                                                                                                                                                                                                                                  | AMB Liposomal / Lipid Complex                                                                                                                                                                                                                                                                                   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A         | I.V.; Intrathecal                                                                                                                                                                                                                                                                                                                                                              | I.V.                                                                                                                                                                                                                                                                                                            |
| D         | Amphotericin B is released from deoxycholate in the bloodstream. <ul style="list-style-type: none"><li>Protein binding 90%</li><li>High concentrations in liver, spleen, kidney, bone marrow, lung</li><li>Lower but sufficient concentrations may be achieved inflamed pleura, synovium, and aqueous humor</li><li>Poor CNS penetration even with inflamed meninges</li></ul> | $V_d$ increases with higher doses (increased uptake by tissues) <ul style="list-style-type: none"><li>Higher concentrations in liver, spleen, and lungs than with conventional AMB<ul style="list-style-type: none"><li>Higher concentration → enhanced effect</li></ul></li><li>Poor CNS penetration</li></ul> |
| H         | Not metabolized<br>Minor excretion in urine<br>No dosage adjustments provided for kidney or liver impairment                                                                                                                                                                                                                                                                   | Not metabolized<br>Minimal excretion in urine<br>No dosage adjustments provided for kidney or liver impairment (not studied)                                                                                                                                                                                    |
| $t_{1/2}$ | Variable   Initial: 15-48 hours   Terminal: 15 days                                                                                                                                                                                                                                                                                                                            | <b>Nonlinear kinetics:</b><br>$t_{1/2_{terminal}}$ 100-173 hours following multiple doses                                                                                                                                                                                                                       |

# Amphotericin B (AMB) Formulations

## Conventional AMB

- Amphotericin B is formulated with sodium deoxycholate (a bile salt), forming a colloidal suspension that enhances AMB's water solubility and stability

## Liposomal and Lipid Complex AMB



- AMB binds the lipids in the lipid vehicles.
- AMB affinity is between that for fungal ergosterol and for human cholesterol
- The liposomes preferentially attach to the fungal cell wall.
- The active AMB molecule is released and transfers to the cell membrane where it can exert its activity that leads to cell death.

**TAKEAWAY:** This slide and the next are to help you conceptualize and understand AMB formulations. They are **NOT** for memorization.

The efficacy of the different formulations appears to be similar.

- Conventional AMB (C-AMB)
- Liposomal AMB (L-AMB)
- AMB lipid complex (ABLC)

Liposomal AMB and AMB lipid complex:

***Infusion-related toxicity and nephrotoxicity are reduced***, which permits use of larger doses compared to conventional AMB.

AMB lipid formulations enhance delivery to the site of infection while minimizing organ toxicity.

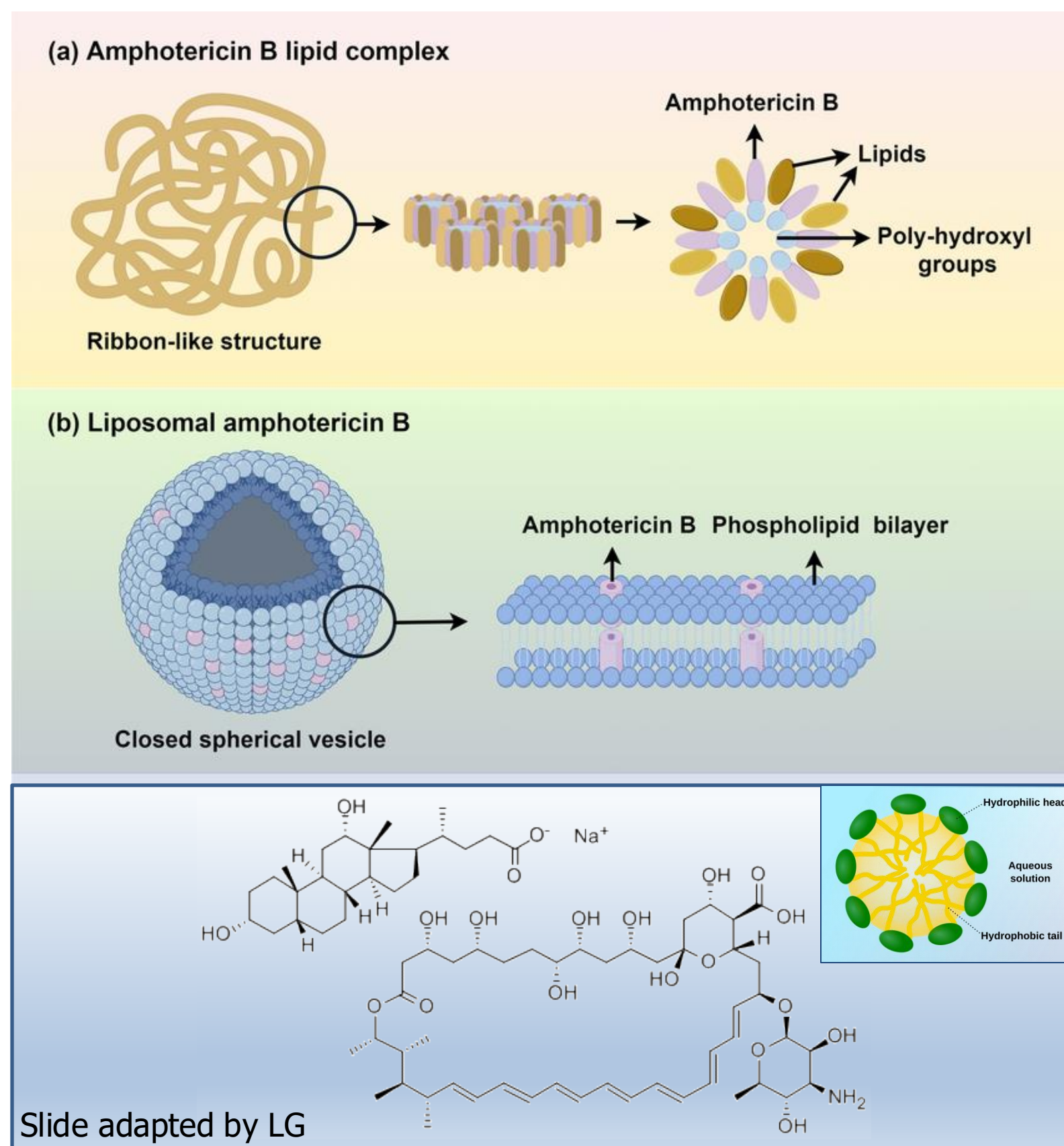
Higher doses can improve effectiveness, particularly in severe or resistant fungal infections.



The putative structure of lipid-based formulations of AmB. (By Figdraw).

- AmB lipid complex (ABLCL) has ribbon-like structures and consists of AmB complexed with two lipids (dimyristoylphosphatidylcholine and dimyristoylphosphatidylglycerol, with a molar ratio of 7:3) in a 1:1 drug-to-lipid molar ratio.
- Liposomal AmB (L-AmB) is a single bilayer liposomal drug delivery system that consists of a mixture of specific proportions of amphophilic substances (e.g., phospholipids and cholesterol) to form closed spherical vesicles.
- AmB deoxycholate consists of amphotericin B complexed with deoxycholate, a bile salt, forming a micellar suspension to enhance solubility.

Figure - available from: Journal of Neurology  
[https://www.researchgate.net/figure/The-putative-structure-of-lipid-based-formulations-of-AmB-By-Figdraw-a-AmB-lipid\\_fig4\\_377811114](https://www.researchgate.net/figure/The-putative-structure-of-lipid-based-formulations-of-AmB-By-Figdraw-a-AmB-lipid_fig4_377811114)  
 Figure micelle: Wikipedia By SuperManu - Own work, CC BY-SA 3.0,  
<https://commons.wikimedia.org/w/index.php?curid=2902736>

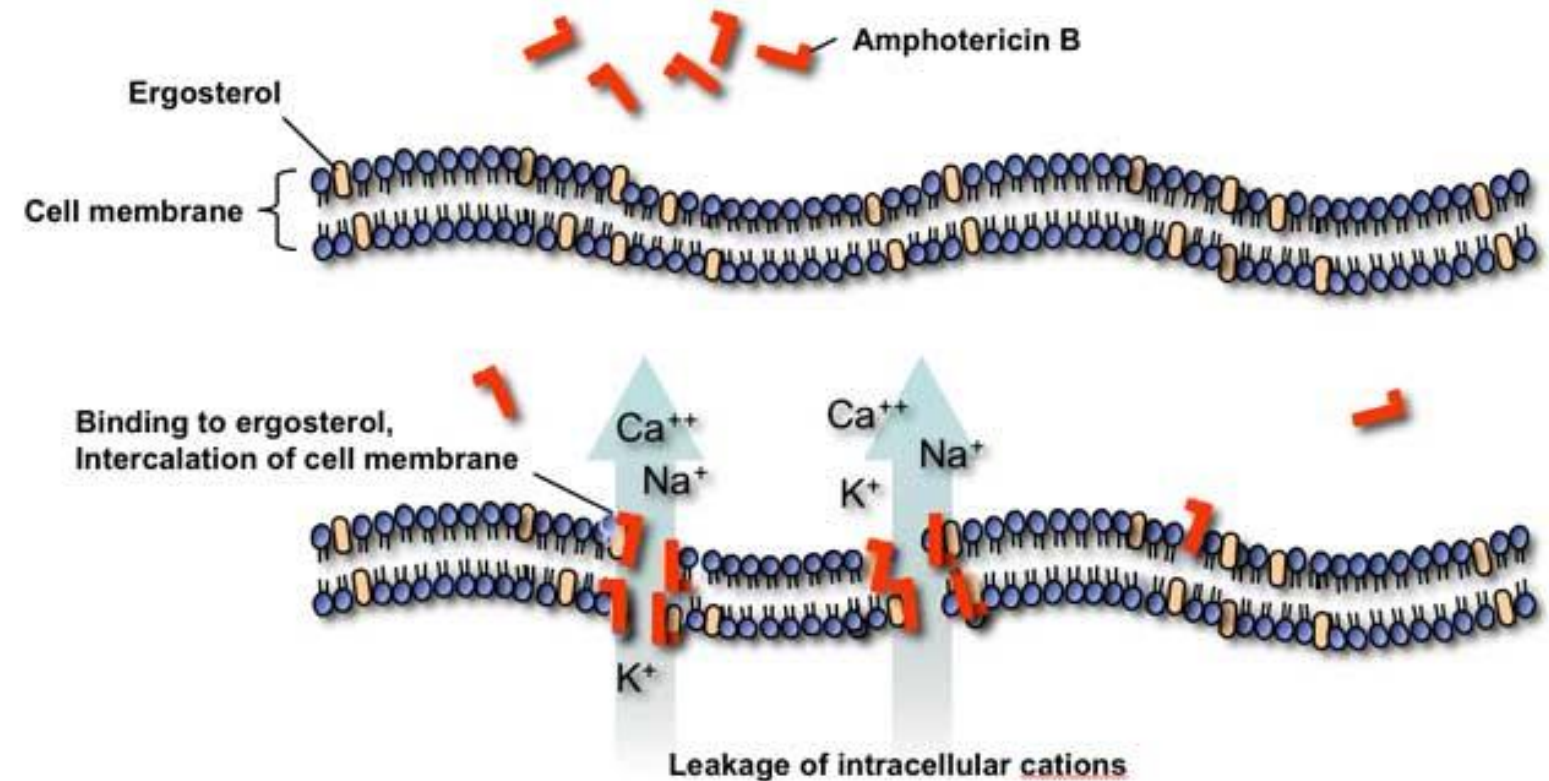


## Amphotericin B Antifungal Action

The **amphipathic characteristic** facilitates AMB's avid combination with ergosterol in the fungal cell membrane along the double bond-rich side of the drug's structure and associates with molecules of the polar heads along the hydroxyl-rich side. The **amphoteric property** (acid/base) enhances binding to ergosterol and helps to sequester ergosterol, further destabilizing the cell membrane.

- (1) AMB associates with ergosterol in fungal membrane  
↓  
forms pores  
↓  
Ions and other small molecules leak out  
*plus*
- (2) Ergosterol is sequestered further compromising membrane integrity  
↓  
fungal cell death

Relative selectivity:  
AMB **binds human membrane sterols**,  
but less avidly than to ergosterol.



<https://drfungus.org/knowledge-base/antifungal-pharmacology/>

### Acquired Resistance (very rare)

- Mutations in various *ERG* genes likely reduces ergosterol concentration in the cell membrane of resistant isolates → reduces susceptibility to AMB.
- Replacement of ergosterol with precursor sterols → reduces susceptibility to AMB.

## Amphotericin B Therapeutic Uses

### Aspergillosis

- Pulmonary
- Extrapulmonary
- Febrile neutropenia
- Allogeneic bone marrow transplant
- Ocular (intraocular admin)

### Mucormycosis

- Rhino-orbital-cerebral and pulmonary infections
- High mortality rates

Cryptococcal meningitis (*C. neoformans*)

Candidiasis, deeply invasive

Some species of *Aspergillus*, *Candida*, *Scedasporium*, *Fusarium*, and causes of chromoblastomycosis are intrinsically resistant.

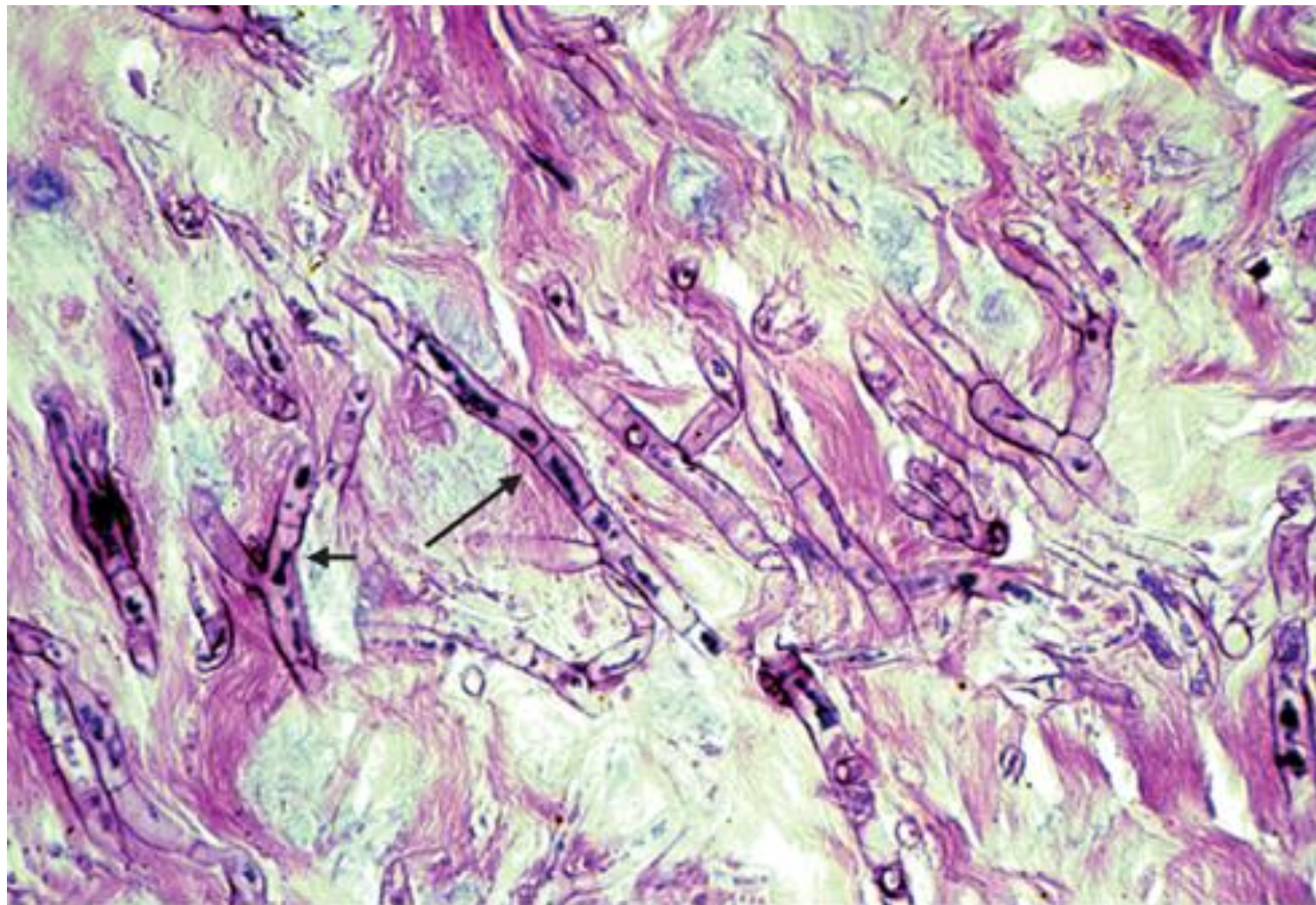
AMB is primarily for treatment of patients with progressive and potentially life-threatening fungal infections.

AMB should **not** be used to treat noninvasive forms of fungal disease such as oral thrush, vaginal candidiasis, and esophageal candidiasis in patients with normal neutrophil counts.

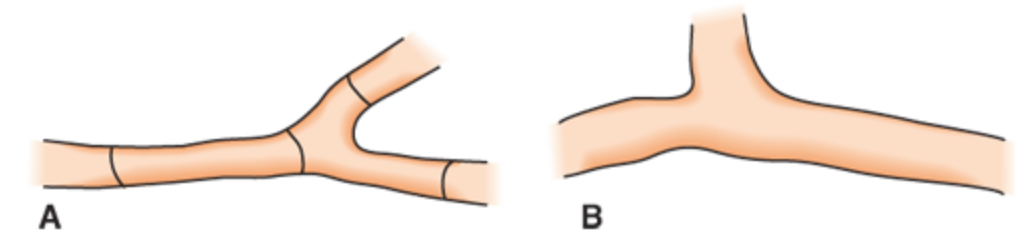
Pregnancy: AMB crosses the placenta and may be found in fetal circulation.

AMB is recommended for treatment of serious fungal infections in pregnancy.





Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz:  
Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious  
Diseases, Fifteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.



Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz:  
Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious  
Diseases, Fifteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

*Aspergillus* and *Mucor* in tissue.

**A:** *Aspergillus* has septate hyphae with V-shaped branching.

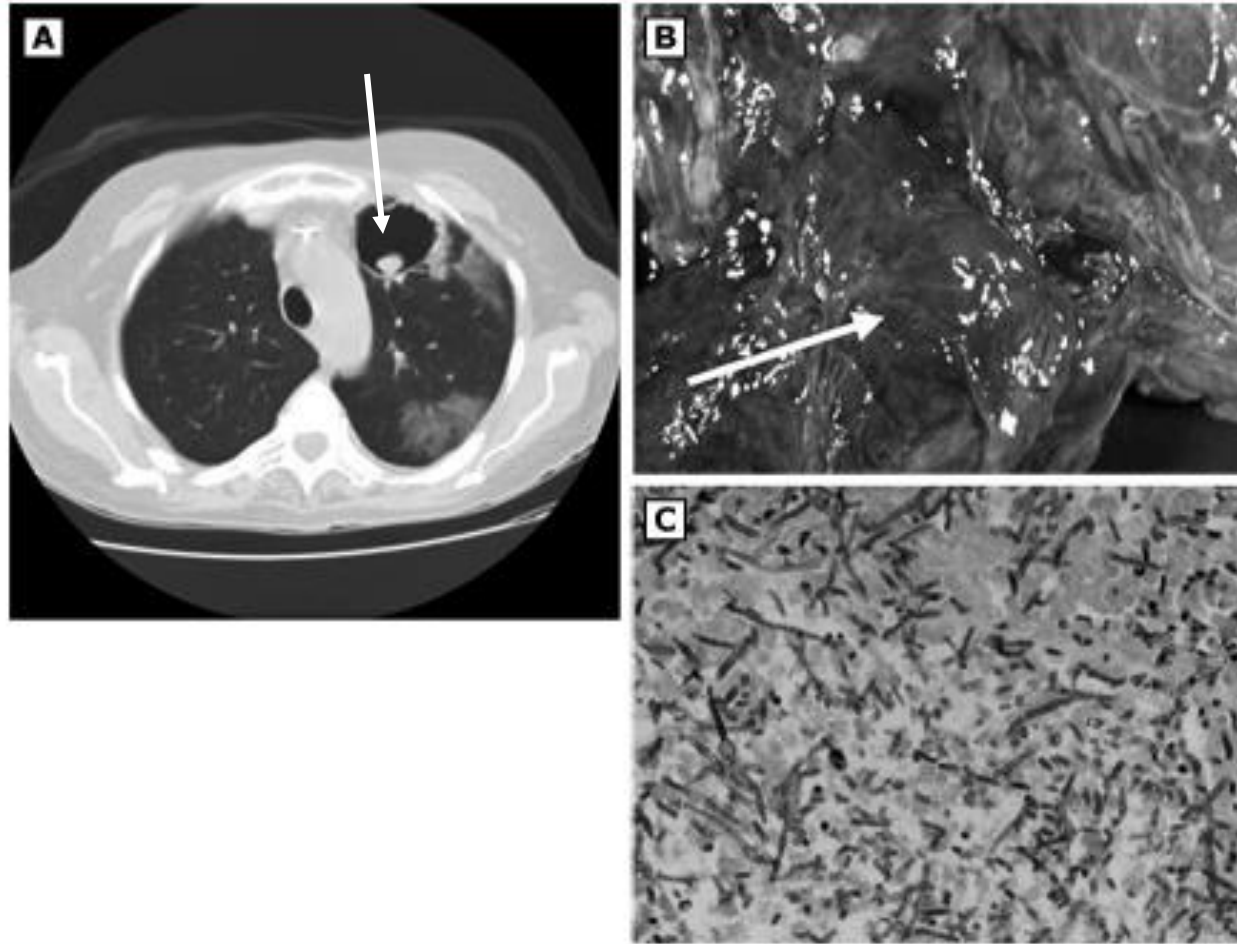
**B:** *Mucor* has nonseptate hyphae with right-angle branching.

Figure 50-9

*Aspergillus fumigatus*—septate hyphae. Long arrow points to the septate hyphae of *Aspergillus*. Note the straight parallel cell walls of this mold. Short arrow points to the typical low-angle, Y-shaped branching. (Used with permission of Prof. Henry Sanchez, University of California, San Francisco School of Medicine.) Figure 50-10



## Invasive aspergillosis in lung transplant



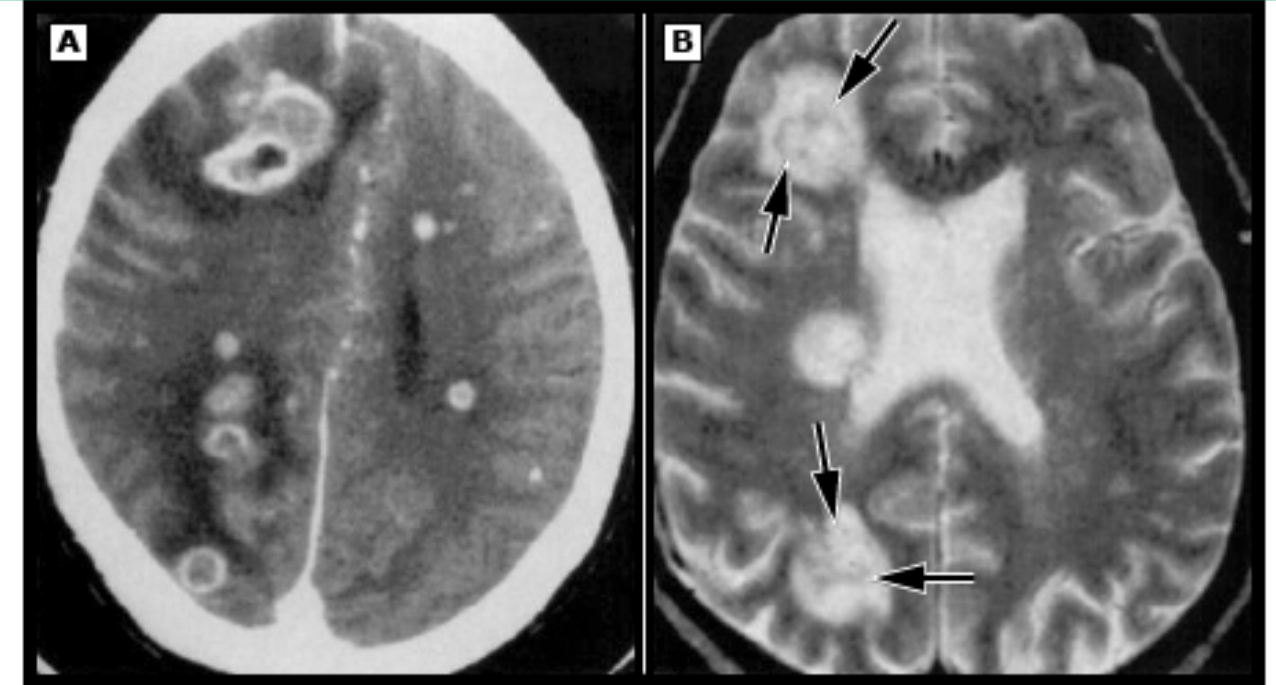
(A) Computed tomography (CT) scan of the chest showing fungal ball with pulmonary infiltrates, (B) gross anatomy on autopsy of the same patient showing mottling of the involved lobe, and (C) Grocott stain of the lung tissue of same patient showing multiple acute angle septate hyphae of *Aspergillus*.

Reproduced with permission from: Silveira FP, Husain S. Fungal infections in lung transplant recipients. *Curr Opin Pulm Med* 2008; 14:211.

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## *Aspergillus* brain abscesses



A 44-year-old man with hairy-cell leukemia and biopsy-proved cerebral aspergillosis.

(A) Computed tomography scan obtained after injection of contrast material shows multiple ring-enhancing lesions with surrounding edema and mass effect.

(B) T2-weighted magnetic resonance image (at slightly lower level than A) shows multiple hypointense rings (arrows) with surrounding edema in right cerebral hemisphere. Rings are irregular and relatively poorly formed.

From: Ashdown BC, Tien RD, Felsberg GJ. *Aspergillosis of the brain and paranasal sinuses in immunocompromised patients: CT and MR imaging findings. AJR Am J Roentgenol* 1994; 162:155. Reprinted with permission from: the American Journal of Roentgenology.

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## Example of Antifungal Treatment Regimen: Treatment of Cryptococcal Meningitis

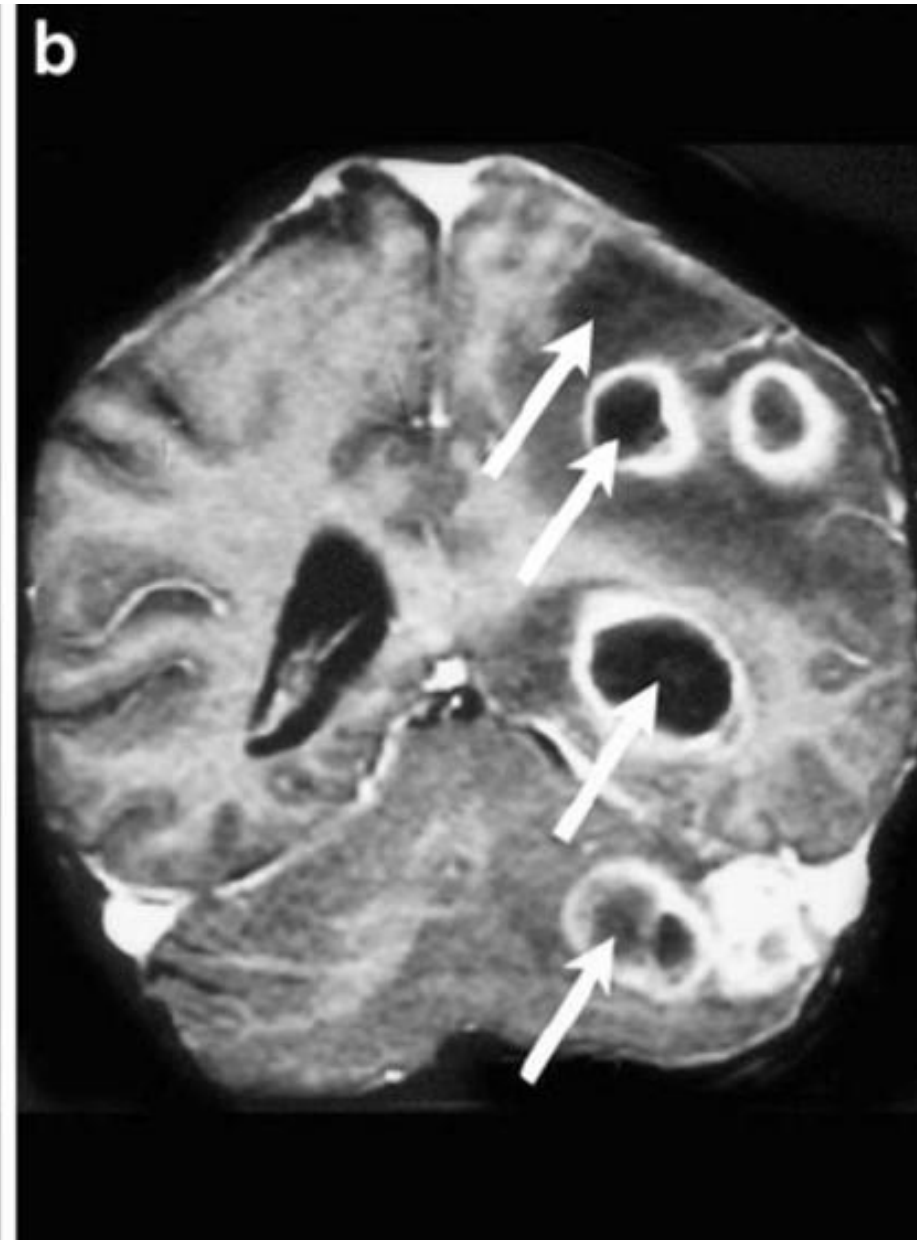
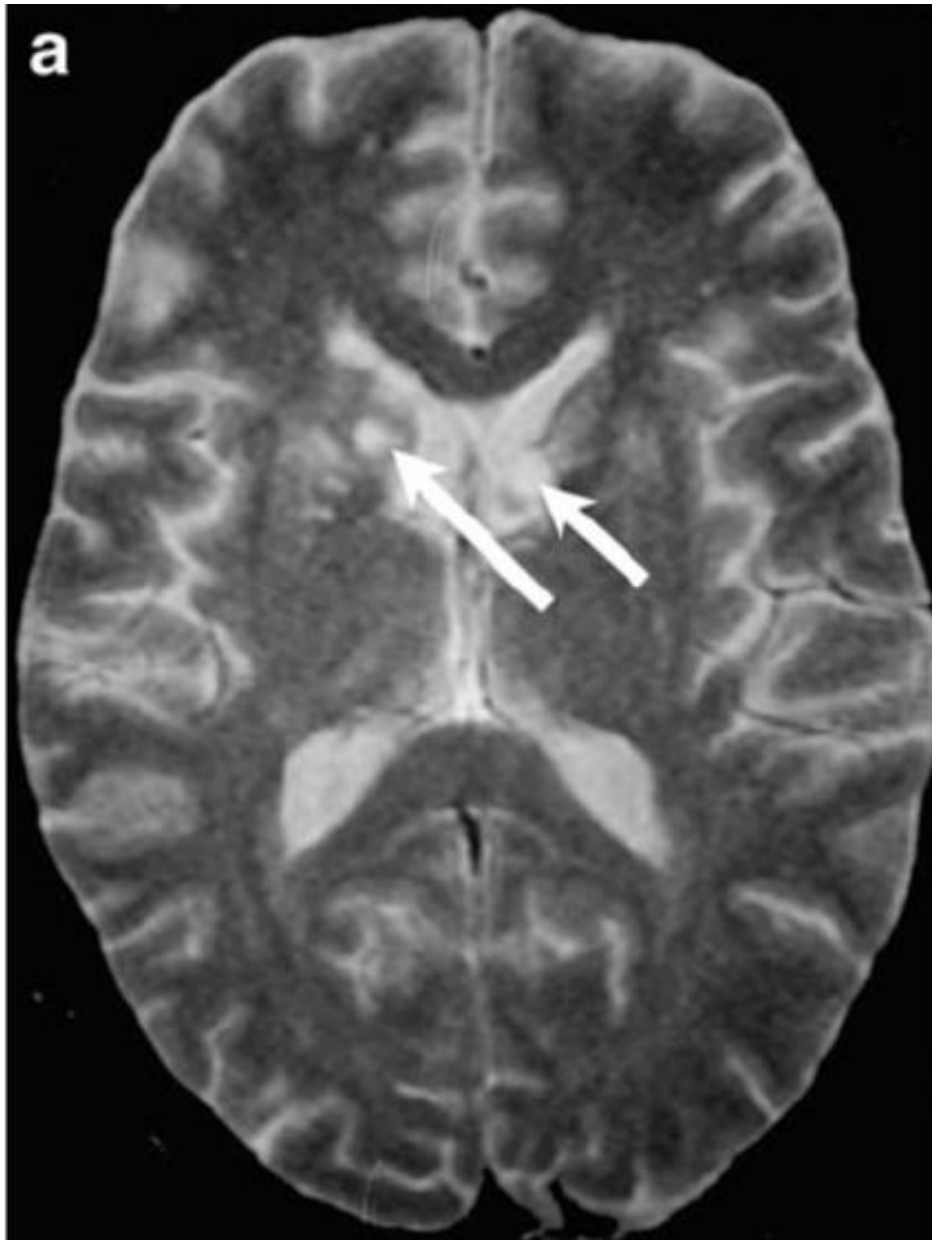
### WHAT YOU NEED TO KNOW:

Induction, Consolidation, Maintenance,  
the drugs being used and why

Amphotericin B: First-line in combination with flucytosine (patients with or without AIDS)

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INDUCTION<br>phase                    | <p>Liposomal AMB or AMB lipid complex I.V. 1x daily<br/>PLUS<br/>Flucytosine (oral) q6h x 2-4 wks or until afebrile and CSF cultures are negative</p> <ul style="list-style-type: none"> <li>– Monitor / manage intracranial pressure</li> <li>– For HIV/AIDS patients, deferring ART for 5 weeks after initiation of therapy significantly improved survival compared to starting ART during the first 2 weeks.</li> </ul> <p style="text-align: right;">ART=Antiretroviral therapy</p> |
| CONSOLIDATION<br>phase                | Fluconazole 400-800 mg 1x daily for 8 weeks (non-AIDS) / 10 weeks (AIDS)                                                                                                                                                                                                                                                                                                                                                                                                                 |
| MAINTENANCE<br>phase<br>(suppression) | <p>Fluconazole 200mg 1x daily to prevent relapse 6-12 months (non-AIDS)</p> <ul style="list-style-type: none"> <li>– For HIV/AIDS patients, discontinuation of therapy may be considered in patients who remain asymptomatic, with CD4 cell count <math>&gt;100</math> cells/mm<sup>3</sup> on ART</li> </ul>                                                                                                                                                                            |

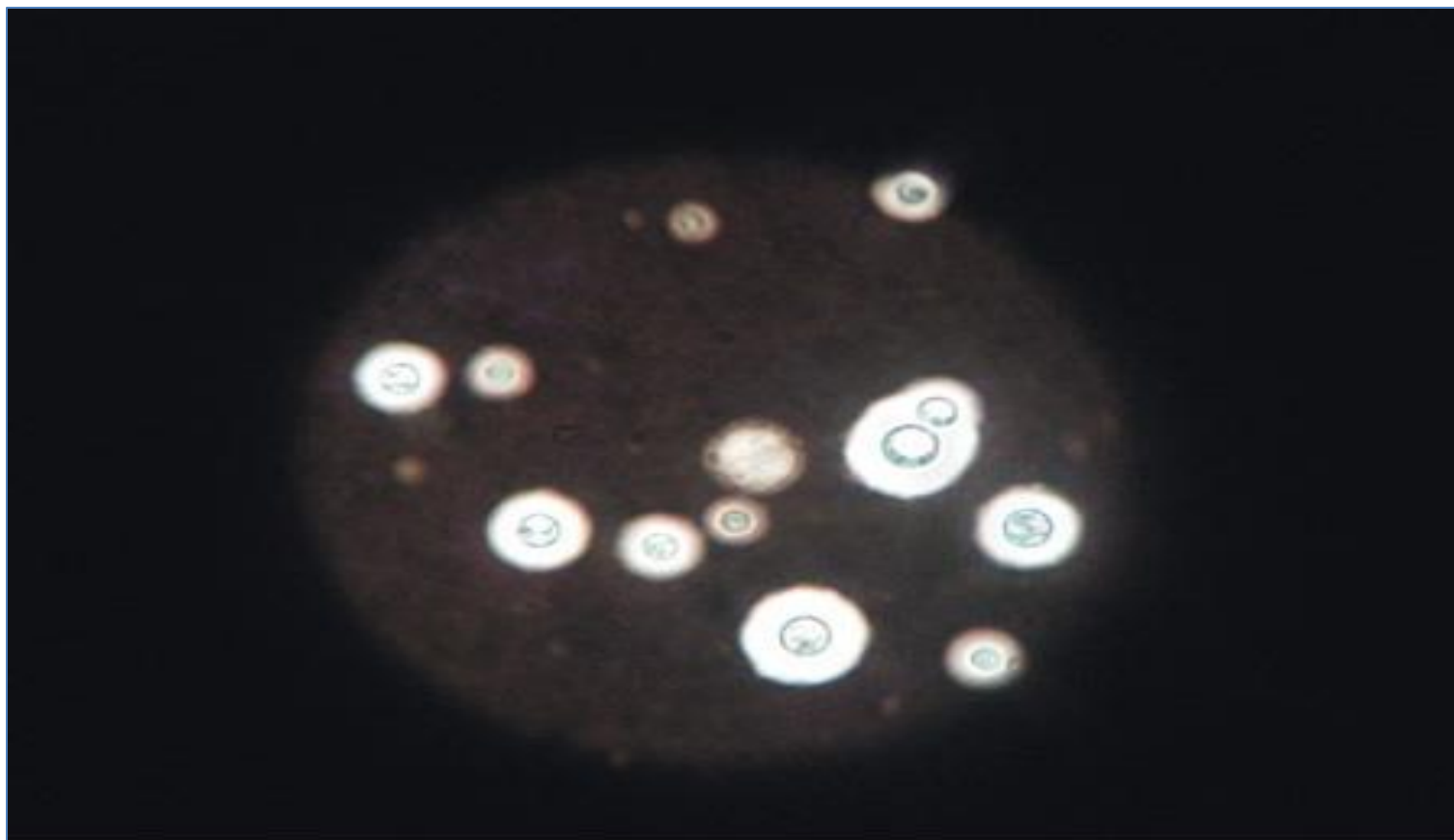




(a) T2-weighted MR image of a patient with cryptococcal meningitis with bilateral infarctions in the caudate nuclei (white arrows).

(b) T1-weighted MR image of a patient with toxoplasmosis.

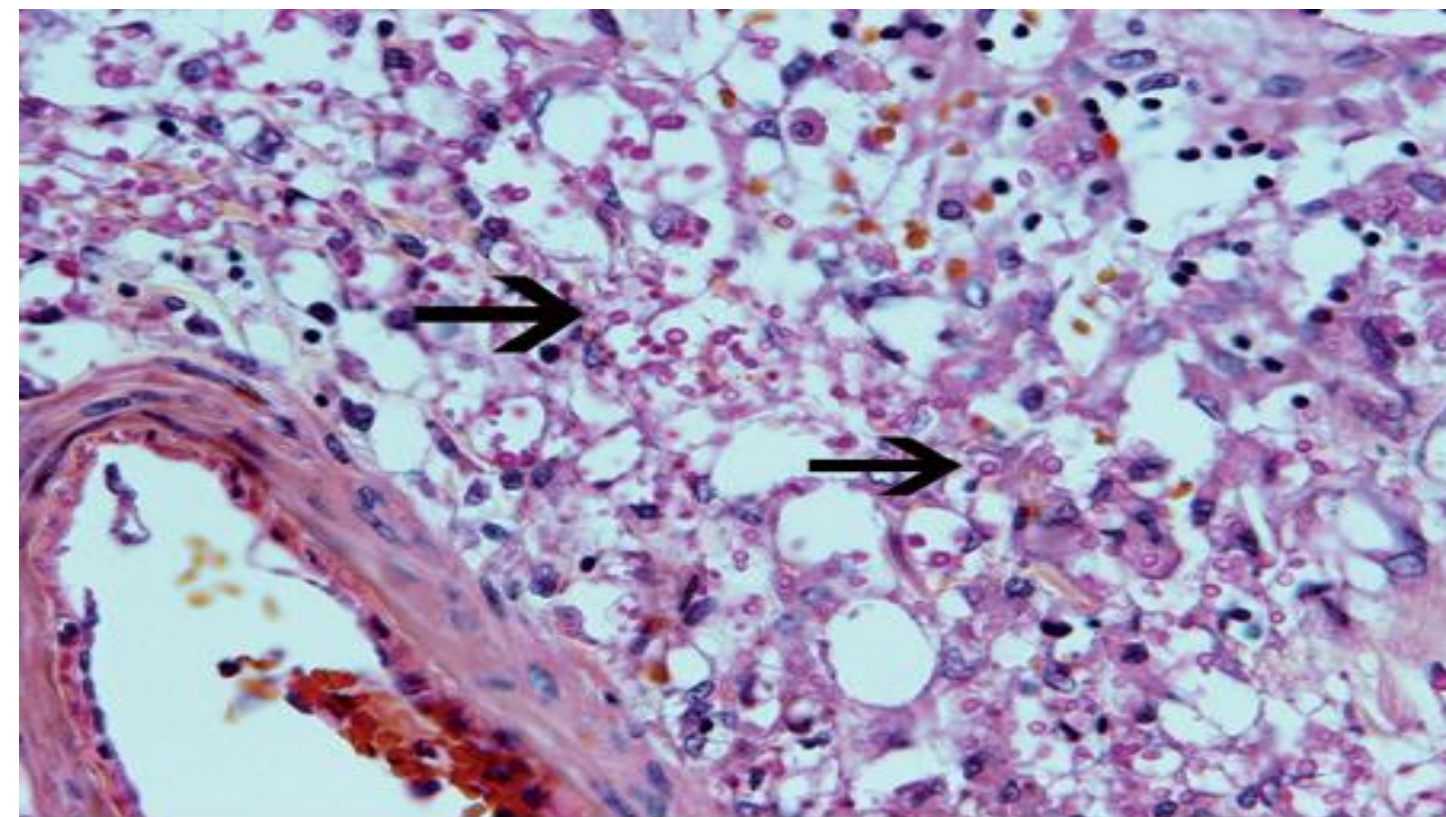
Arrows illustrate the frank lesions as well as edema (dark hypointense area surrounding the bright ringed lesion) resulting from injury.



Source: Knoop KJ, Stack LB, Storrow AB, Thurman RJ: *The Atlas of Emergency Medicine, 3rd Edition*: <http://www.accessmedicine.com>  
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*Cryptococcus neoformans* seen as encapsulated yeast on India ink preparation of the CSF from an HIV patient with cryptococcal meningitis.

The Atlas of Emergency Medicine, 3e, 2010; Figure 20-17  
(Photo contributor: Seth W. Wright, MD.)



Source: Kemp WL, Burns DK, Brown TG: *Pathology: The Big Picture*: [www.accessmedicine.com](http://www.accessmedicine.com)

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Cryptococcal meningitis.

This high-power photomicrograph of the meninges, stained with mucicarmine, highlights encapsulated and budding yeast forms.

The capsule of *Cryptococcus neoformans* stains with mucicarmine (*arrows*).

Pathology, The Big Picture Figure, 2008; Figure 11-24  
(Photo contributor: Seth W. Wright, MD.)

**AMB therapy is often limited by toxicity, especially drug-induced renal impairment.**

➤ AMB lipid formulations reduce infusion reactions and nephrotoxicity.

| Toxicity         | Comments                                                                                                                                                                                                                                                        |                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Nephrotoxicity   | Azotemia                                                                                                                                                                                                                                                        | High serum levels of nitrogenous wastes products occurs in nearly all patients. |
|                  | Acute kidney injury                                                                                                                                                                                                                                             | ↓ renal perfusion → ↓ glomerular filtration rate (GFR)                          |
|                  | Renal tubular acidosis                                                                                                                                                                                                                                          | K <sup>+</sup> and Mg <sup>2+</sup> wasting occur                               |
|                  | Hydration with normal saline infusions is recommended.                                                                                                                                                                                                          |                                                                                 |
| Hematologic      | • Anemia                                                                                                                                                                                                                                                        | ↓ erythropoietin production (by specialized kidney cells)                       |
| Infusion-related | <ul style="list-style-type: none"><li>• Fever, chills, rigor, vomiting, headache, malaise, hypotension, tachypnea</li><li>• Thrombophlebitis</li><li>• Premedication with acetaminophen, antihistamine, glucocorticoicoid, meperidine may be helpful.</li></ul> |                                                                                 |
| Intrathecal      | • Headache, fever common; arachnoiditis, delirium, seizures possible                                                                                                                                                                                            |                                                                                 |



## Check your knowledge

What are the advantages of liposomal AMB and AMB lipid complex compared to AMB deoxycholate?

What is AMB's mechanism of antifungal action?

What are the toxicities associated with AMB administration?



## Check your knowledge

What physical properties of the amphotericin B molecule are probably important for its fungus-killing activity?

What accounts for its relatively selective toxicity against fungi?

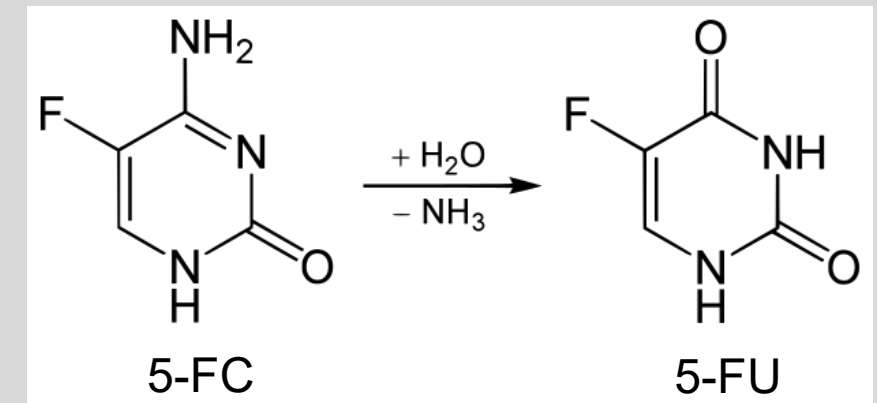
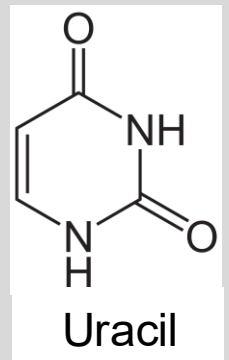
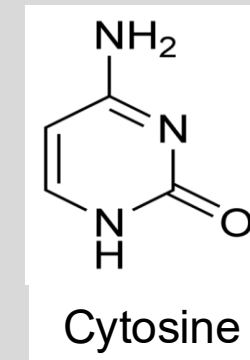
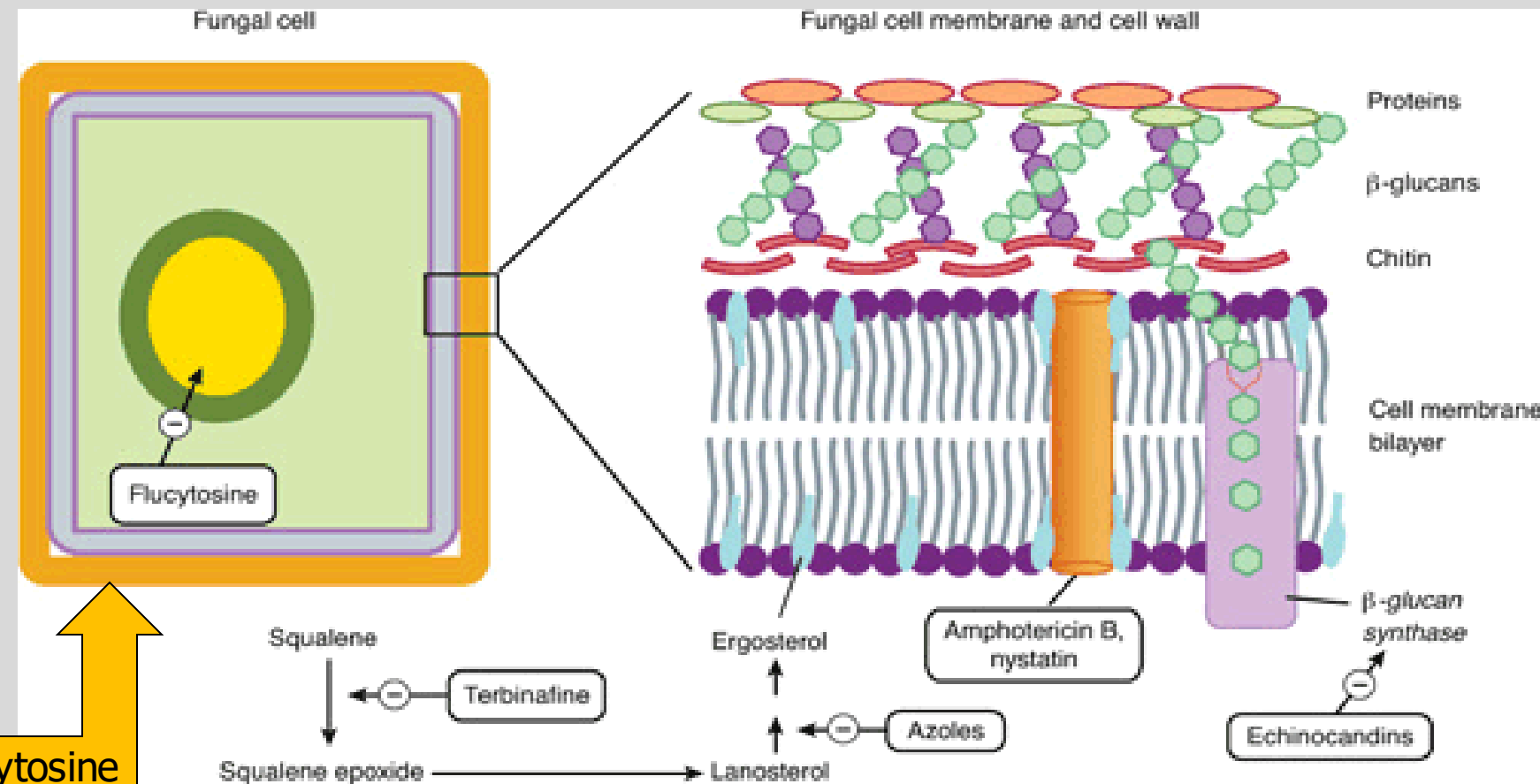
What are the likely mechanisms of resistance?

AMB is an important therapeutic for what patients and infections?

# \*Flucytosine (5-FC)

fluorinated pyrimidine analog

converted to 5-fluorouracil (5-FU) in fungal cells

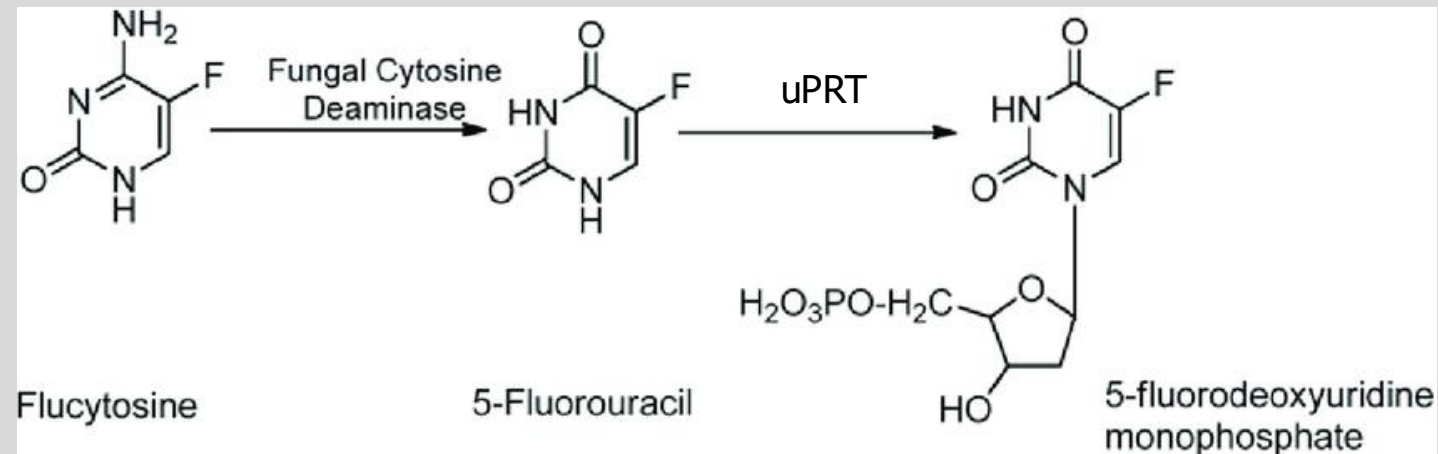


## Flucytosine (5-FC) Pharmacokinetics

|                 |                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PK</b>       | Oral (prodrug); rapid, good absorption, bioavailability ~90%, <b>good CSF penetration</b> , intracellular deamination in yeasts and certain gut bacteria to active 5-FU; urine excretion of 5-FC ( <b>renal dosing adjustments are necessary</b> ); $t_{1/2}$ 3-6h, up to 200h in renal failure; admin q6h                                                                  |                                                                                                                                                                                                                |
| <b>MOA</b>      | Inhibits RNA and DNA synthesis ( <i>See next slide.</i> )<br><b>Resistance emerges rapidly</b> during therapy when the drug is used alone. <ul style="list-style-type: none"> <li>• Loss of cytosine permease transporter</li> <li>• ↓ Cytosine deaminase activity or uPRT activity</li> </ul>                                                                              | <ul style="list-style-type: none"> <li>• AMB is nephrotoxic in many patients.</li> <li>• 5-FC dose must be adjusted in renal insufficiency.</li> <li>• 5-FC toxicities are concentration-dependent.</li> </ul> |
| <b>Activity</b> | Fungistatic<br>Narrow spectrum: <i>Cryptococcus neoformans</i> , <i>Candida</i> some species, most fungi that cause chromoblastomycosis, and some dematiaceous (darkly pigmented) fungi                                                                                                                                                                                     |                                                                                                                                                                                                                |
| <b>Uses</b>     | <i>Cryptococcus</i> meningitis and selected invasive candidiasis<br>5-FC has additive or synergistic effect with AMB (in vitro testing).                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |
| <b>AE</b>       | Well tolerated when renal function is normal<br>Concentration-dependent adverse effects: <ul style="list-style-type: none"> <li>• Heme: leukopenia, thrombocytopenia</li> <li>• Liver: elevated liver function enzymes (AST/ALT); liver necrosis reported</li> <li>• GI: nausea, vomiting diarrhea</li> </ul> Pregnancy: adverse events seen in animal reproduction studies | <p>Certain bacteria in the gut microbiome convert 5-FC to 5-FU, which is absorbed and <b>can cause host cell toxicity.</b></p>                                                                                 |

## Flucytosine Mechanism of Action

- **Entry:** fungal cell uptake via **cytosine permeases**
- **Activation:** Deamination in fungal cell to 5-FC → 5-FU

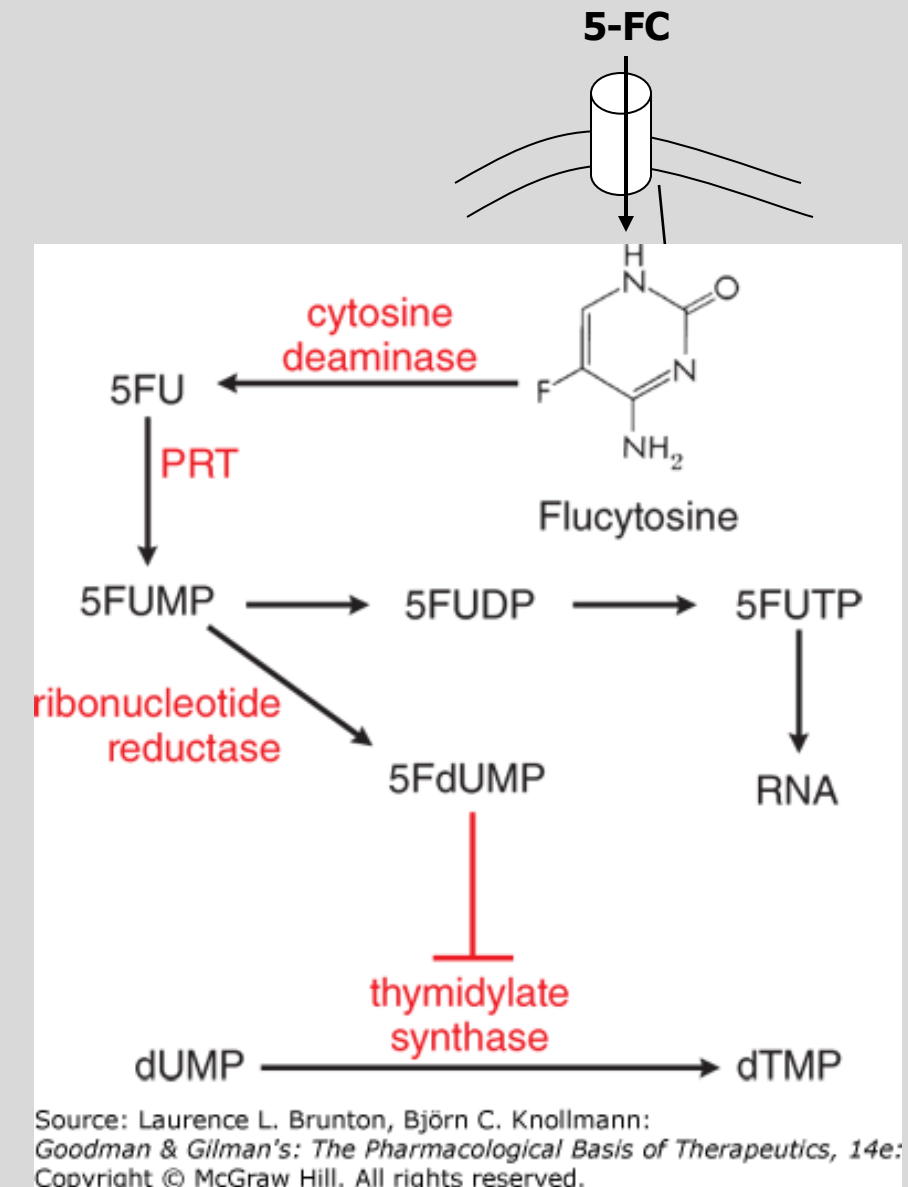


### Antifungal action: Activation by cellular kinases

1. **RNA:** 5-FUTP incorporation into RNA → reduces RNA synthesis and, therefore, protein synthesis
2. **DNA:** 5-FUMP → 5-FdUMP → potent thymidylate synthase inhibition → reduces DNA synthesis

### Selectivity for fungal cell:

Human cells lack cytosine permease, cytosine deaminase, and uracil phosphoribosyltransferase (uPRT) activity.





## Check your knowledge

What is the basis of flucytosine's selectivity?

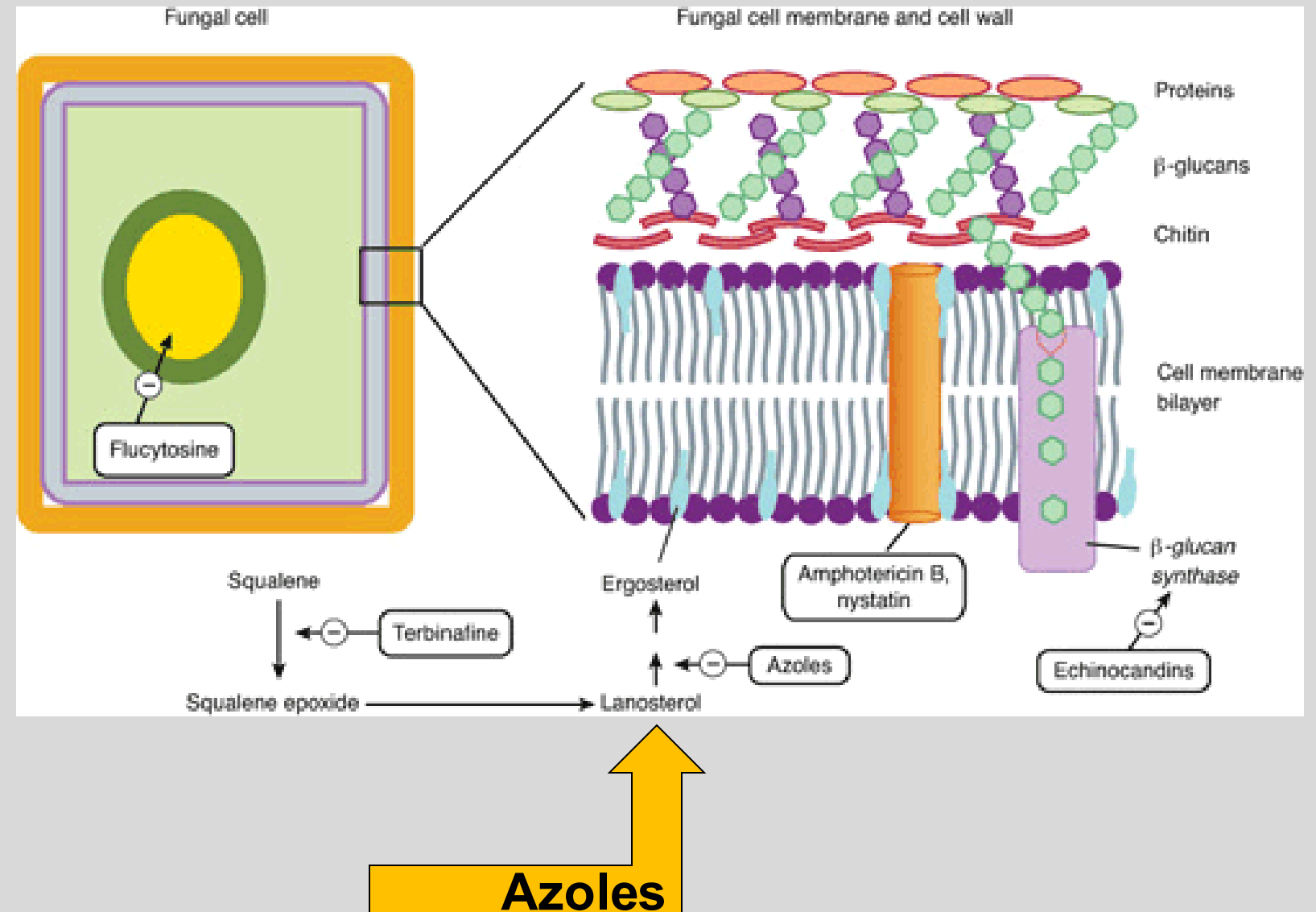
What is the basis of flucytosine's toxicity?

How does flucytosine induce cellular cytotoxicity?

What are the therapeutic uses of 5-FC and why?

What are major adverse effects of 5-FC and how can they be reduced?

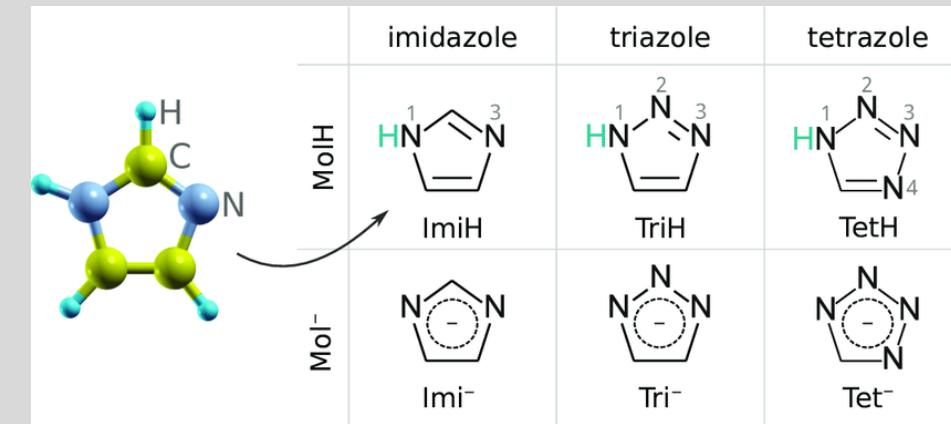
# Azole Antifungals



# Azole Antifungal Agents

## Broad spectrum of activity | Fungistatic

Some azoles appear to fungicidal against *Aspergillus* spp.



### For Systemic Use

Triazoles

- \*Itraconazole
- \*Fluconazole
- \*Voriconazole
- \*Posiconazole
- \*Isavuconazole (Prodrug: Isavuconazonium)

#### Oteseconazole

Oteseconazole is a tetrazole antifungal drug for oral treatment of recurrent vulvovaginal candidiasis.

#### \*Ketoconazole

Because of its toxicity, oral ketoconazole has been replaced by systemic triazole antifungals.

### Topical Products

- \*Clotrimazole
- \*Ketoconazole
- Butoconazole
- Miconazole
- Econazole
- Luliconazole
- Oxiconazole
- Sertaconazole
- Sulconazole
- Tiaconazole
- Terconazole
- Efinaconazole

Imidazoles

## Pharmacokinetics of Azole Antifungal Agents

**TAKEAWAYS:** Low protein binding → diffusion across BBB. Bioavailability and  $t_{1/2}$  vary. Extensive hepatic metabolism (most drugs). Inhibition of CYP3A4 (and others).

|               | Route<br>Bioavailability | CSF<br>Penetration | Elimination / substrate<br><b>CYP inhibition</b>                                   | $t_{1/2}$<br>(hours) | Drug<br>Interactions                        |
|---------------|--------------------------|--------------------|------------------------------------------------------------------------------------|----------------------|---------------------------------------------|
| Itraconazole  | Oral<br>Variable*        | No                 | Hepatic / CYP3A4<br><b>CYP3A4, P-gp</b>                                            | 24-42                | <b>High potential for drug interactions</b> |
| Fluconazole   | Oral, IV<br>>90%         | Yes                | <b>Renal</b> / minor hepatic<br>~80% unchanged (urine)<br><b>CYP2C19, 2C9, 3A4</b> | 22-31                |                                             |
| Voriconazole  | Oral, IV<br>96%          | Yes                | Hepatic CYP2C19, 2C9, 3A4<br><b>CYP3A4, 2C9</b>                                    | 6                    |                                             |
| Posaconazole  | Oral, IV<br>Variable*    | No                 | Minimal (~17% by UGT)<br>excreted unchanged (feces)<br><b>CYP3A4, P-gp</b>         | 25                   |                                             |
| Isavuconazole | Oral, IV<br>98%          | limited<br>data    | Hepatic / CYP3A4, UGT<br><b>CYP3A4, P-gp</b>                                       | 130                  |                                             |
| Ketoconazole  | Oral<br>Variable*        | No                 | Hepatic<br><b>CYP3A4, P-gp</b>                                                     | 7-10                 |                                             |

\*Itraconazole, Posaconazole, Ketoconazole: acid environment for optimal absorption

**Bold:** strong inhibitors



## Azole antifungal action:

### Inhibition of 14- $\alpha$ -demethylase inhibits ergosterol biosynthesis

14- $\alpha$  demethylase (CYP51) is an enzyme expressed by *ERG11* gene.  
Azole antifungals bind the iron in the heme group.

squalene  $\rightarrow$   $\rightarrow$  lanosterol  $\xrightarrow{14\text{-}\alpha\text{-demethylase}}$  ...  $\rightarrow$  ergosterol

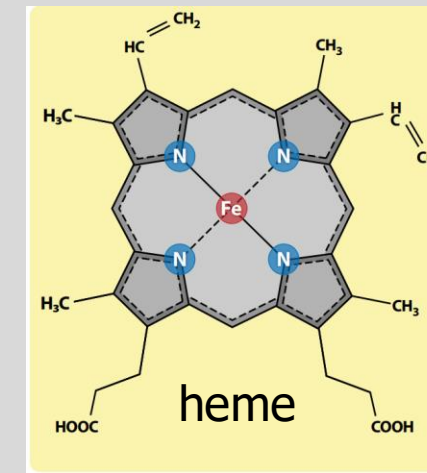
**azole antifungals**

Azole antifungals block conversion of lanosterol to ergosterol, which decreases ergosterol concentration in fungal cell membranes, inhibiting cell membrane formation and

Accumulation of toxic 14-methylated sterols (intermediates) disrupts the integrity and function of the fungal cell membrane.

**growth arrest**

At high concentrations, as can be achieved with topical application, some azoles directly increase fungal plasma membrane permeability.



### Resistance:

1. Modifications of target enzymes  
Mutation in *ERG11* gene  $\rightarrow$   $\downarrow$  binding of the azole to 14- $\alpha$ -demethylase and allows access of the natural substrate, lanosterol (observed in *C. albicans*).
2. Efflux pumps transport the azole out of the fungal cell.
3.  $\uparrow$  Production of lanosterol, which outcompetes the drug at the catalytic site.

## Summary of azole antifungals spectrums of activity (susceptible fungi)

|                                                                                                                                                                                                                          | Itraconazole | Fluconazole | Voriconazole | Posaconazole | Isavuconazole |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------|--------------|---------------|
| Aspergillus                                                                                                                                                                                                              | ✓            | —           | ✓            | ✓            | ✓             |
| Candidiasis                                                                                                                                                                                                              | ✓            | ✓           | ✓            | ✓            | ✓             |
| C. neoformans                                                                                                                                                                                                            | ✓            | ✓           | ✓            | ✓            | ✓             |
| Mucormycosis                                                                                                                                                                                                             | —            | —           | —            | ✓            | ✓             |
| Dimorphic fungi                                                                                                                                                                                                          | ✓            | ±           | ✓            | **           | **            |
| Onychomycosis                                                                                                                                                                                                            | ✓            | ±           | **           | **           | **            |
| Tinea                                                                                                                                                                                                                    | ✓            | ✓           | **           | **           | **            |
| <p>— No activity</p> <p>± Less effective for these fungal infections than other options.</p> <p>** Effective but reserved for infections that can't be treated with standard therapy.</p> <p>Tinea = Dermatophytosis</p> |              |             |              |              |               |

**TAKEAWAY:** The spectrum of action directly influences usefulness in treatment specific infections. Remember to think about risks of resistance and adverse effects for the individual patient.

# Azole Antifungals Therapeutic Uses (susceptible fungi)

**TAKEAWAYS:**  
Spectrums of action→uses, drug of choice, CNS penetration

| Itraconazole                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fluconazole                                                                                                                                                                                                                                                                                       | Voriconazole                                                                                                                                                                                                                                                                                                                         | Posaconazole                                                                                                                                                                                                                                        | Isavuconazole                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Broad spectrum                                                                                                                                                                                                                                                                                                                                                                                                                                      | Good CNS penetration                                                                                                                                                                                                                                                                              | Good CNS penetration                                                                                                                                                                                                                                                                                                                 | <b>Broadest spectrums</b> of the azole antifungals. Generally reserved for patients refractory to or intolerant of first-line drugs.                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Drug or choice for dimorphic fungi</b> <ul style="list-style-type: none"><li>• Blastomycosis</li><li>• Sporotrichosis</li><li>• Histoplasmosis</li><li>• Coccidiomycosis</li><li>• Paracoccidioidomycosis</li></ul> Alternative for: <ul style="list-style-type: none"><li>• Onychomycosis (second to terbinafine)</li><li>• Tinea infections refractory to topical antifungal treatment</li><li>• Aspergillosis</li><li>• Candidiasis</li></ul> | <b>Initial choice for</b> <ul style="list-style-type: none"><li>• Cryptococcal meningitis</li><li>• <b>consolidation and maintenance phases</b></li><li>• Candidemia</li><li>• Mucocutaneous candidiasis</li><li>• Coccidioidomycosis</li></ul> No activity against aspergillosis or mucormycosis | <b>Drug of choice for invasive infections</b> <ul style="list-style-type: none"><li>• Aspergillosis</li><li>• Candidemia, including fluconazole-resistant</li><li>• Disseminated <i>Candida</i> infections</li><li>• Cryptococcal meningitis (alternative for fluconazole)</li><li>• Serious infections by dimorphic fungi</li></ul> | <b>Treatment of invasive infections</b> <ul style="list-style-type: none"><li>• Aspergillosis</li><li>• Mucormycosis, oral step down therapy</li><li>• Oropharyngeal candidiasis</li><li>• Salvage therapy for invasive fungal infections</li></ul> | <b>Treatment of invasive infections</b> <ul style="list-style-type: none"><li>• Aspergillosis</li><li>• Mucormycosis</li></ul> Alternative for <ul style="list-style-type: none"><li>• Cryptococcal meningitis (maintenance and consolidation)</li></ul> <b>ISA penetrates brain parenchyma.</b> <ul style="list-style-type: none"><li>• Esophageal candidiasis</li></ul> |

Posaconazole and isavuconazole are the only azoles with in vitro activity against Mucorales.

Itraconazole, voriconazole, posaconazole and isavuconazole are also used for prophylaxis of invasive fungal infections in hematologic malignancy, hematopoietic stem cell transplant, and solid organ transplant patients.

## Azole antifungals: Class adverse effects

**Monitor for signs and symptoms; Monitor blood levels as appropriate.**

| System                                     | Class effects unless otherwise indicated (not a complete list)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                 |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| GI                                         | The most common side effect: Nausea, abdominal pain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                 |
| Hepatotoxicity                             | Ranges from asymptomatic increase in serum hepatic enzymes (AST/ALT) to liver failure (rare)                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                 |
| Cardiac                                    | Prolongation of QT interval → torsades de pointes → ventricular arrhythmias.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                 |
| Neurological                               | Peripheral neuropathy<br>Onset: 1 to 18 months                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Most common with itraconazole<br>Also occurs with voriconazole and posaconazole |
| Hypersensitivity reactions                 | Immediate: urticaria, angioedema, anaphylactic shock<br>Delayed: Range from rash to severe cutaneous reactions (SJS/TENS)<br>Serum sickness, and others<br>Photosensitivity                                                                                                                                                                                                                                                                                                                                                              |                                                                                 |
| Pregnancy                                  | <b>Avoid – may cause fetal harm. Use only if benefit to mother outweighs risk to fetus.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                 |
| Drug interactions<br><b>High potential</b> | <ul style="list-style-type: none"> <li><b>CYP3A4-related:</b> Co-administration with many drugs metabolized by CYP3A4 is <b>contraindicated</b>.</li> <li><b>QT interval prolongation:</b> Drugs that prolong QT interval or patients with long QT syndrome may cause torsades de pointes (ventricular arrhythmias)</li> <li><b>pH-related:</b> Ketoconazole, itraconazole caps, posaconazole require acidic environment (with food or acidic carbonated beverage, eg, ginger ale); some should be taken on an empty stomach.</li> </ul> |                                                                                 |

Voriconazole and posaconazole IV formulated with cyclodextrin can accumulate in renal insufficiency and acute kidney injury.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Itraconazole</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Congestive heart failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 | Itraconazole is contraindicated in patients with heart failure. |
| <b>Voriconazole</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Ophthalmic effects: Blurred vision, optic neuritis, vision color changes, scleritis</li> <li>Peripheral neuropathy reported: May be rapid, debilitating, irreversible</li> <li>Encephalopathy, hallucinations may occur with IV administration (high concentrations)</li> <li>Periosteal disease: Voriconazole is trifluorinated; F<sup>-</sup> is integrated into extracellular matrix.</li> </ul> |                                                                 |
| <b>Metabolic-related effects: Proposed mechanisms</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |
| <b>Adrenal suppression</b> <ul style="list-style-type: none"> <li><b>Ketoconazole:</b> strong inhibitor of steroidogenesis</li> <li><b>Itraconazole and posaconazole:</b> Effects on steroidogenesis are milder and occur less frequently.</li> </ul> <p>Inhibition of 17-alpha hydroxylase (CYP17) and C17,20 desmolase suppresses steroidogenesis.</p> <ul style="list-style-type: none"> <li>Suppression of androgen synthesis leads to oligospermia, gynecomastia, menstrual irregularities (estrogens are derived from androgens), and decreased libido.</li> </ul> <p>Inhibition of 11-β hydroxylase (CYP11A1) decreases glucocorticoid and mineralocorticoid synthesis.</p> <ul style="list-style-type: none"> <li>Cortisol and aldosterone synthesis are suppressed</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |
| <b>Pseudohyperaldosteronism with hypertension and hypokalemia:</b> <p><b>Itraconazole and posaconazole:</b> More frequently observed</p> <p>Azoles may inhibit the enzyme that inactivates cortisol, 11-β-hydroxysteroid dehydrogenase type 2, which is found in aldosterone-sensitive tissues.</p> <p>Like aldosterone, cortisol activates mineralocorticoid receptors → sodium and water reabsorption, which raises blood pressure, and causes potassium loss.</p>                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |
| Fluconazole, voriconazole, isavuconazole: Low potential for adrenal suppression and pseudohyperaldosteronism.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |

- **When prescribing oral azole antifungal therapy, certain precautions should be considered and recommendations should be made to the patient.**
  1. The patient's concomitant drugs and comorbid conditions in view of the high potential for drug interactions.
  2. The patient's risk of ventricular fibrillation in view of the drug's QT interval prolonging effects.
  3. Food?: Take with food (for gastric acidity) or take on empty stomach?
  4. Patient education about the symptoms and signs of potential side effects (class and drug-specific effects) and hypersensitivity reactions.
  5. The importance of monitoring: liver function studies and therapeutic drug levels as appropriate.

## Check your knowledge

Which drug is first line for treatment of fungal infections by dimorphic fungi?

Which drug is first line for consolidation and maintenance of cryptococcal meningitis?

Which drug is the preferred initial treatment for invasive aspergillosis?

Which drug is an alternative for the initial treatment of invasive aspergillosis in a patient with optic neuritis or peripheral neuropathy?

Which drugs are used as step-down therapy for mucormycosis?

Which oral drug is first line for vulvovaginal candidiasis?

## Check your knowledge

What adverse effects can occur with all of the azole antifungals?

Which azole antifungal should be avoided in patients with a history of heart failure?

Which azole antifungal is associated with optic neuritis and other ocular toxicities?

Which azole antifungal is associated with peripheral neuropathy?

Which azole antifungal is most likely to interfere with steroidogenesis?

## Check your knowledge

What is the major mechanism of drug interactions with the azole antifungal drugs?

One of the azole antifungals is cleared renally and excreted unchanged in urine. Which one? What do we need to think about in terms of dosing this drug?

Two triazole antifungal drugs are formulated with cyclodextrin to solubilize the hydrophobic drugs for intravenous administration. What toxicity can occur in patients with renal insufficiency? What are the two drugs?

One of the antifungals in this group does not require dosage adjustment for patients with renal insufficiency or liver impairment. What is it? Why are dosage adjustments not required?



# Echinocandins

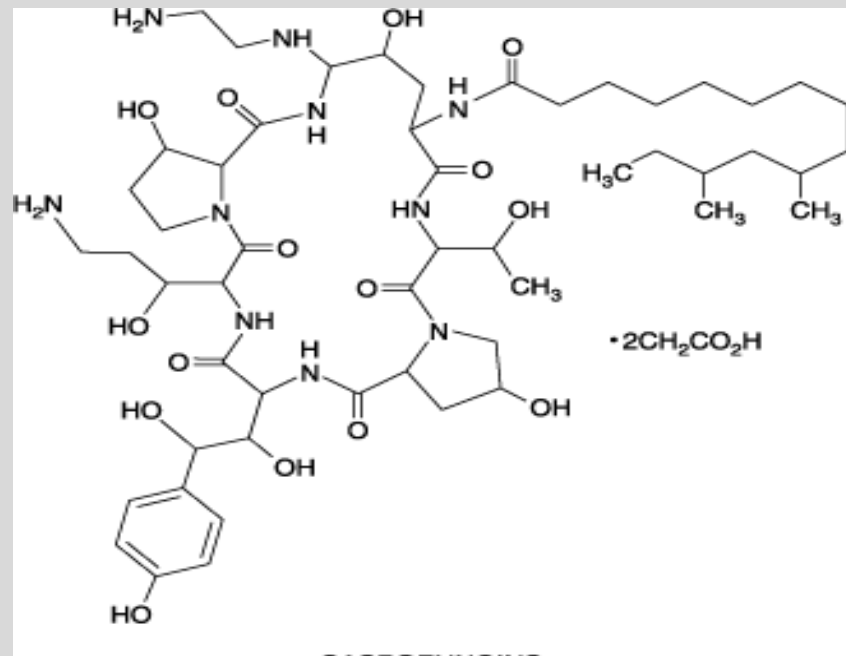
$\beta$ -1,3-glucan synthase inhibitors

\*Caspofungin

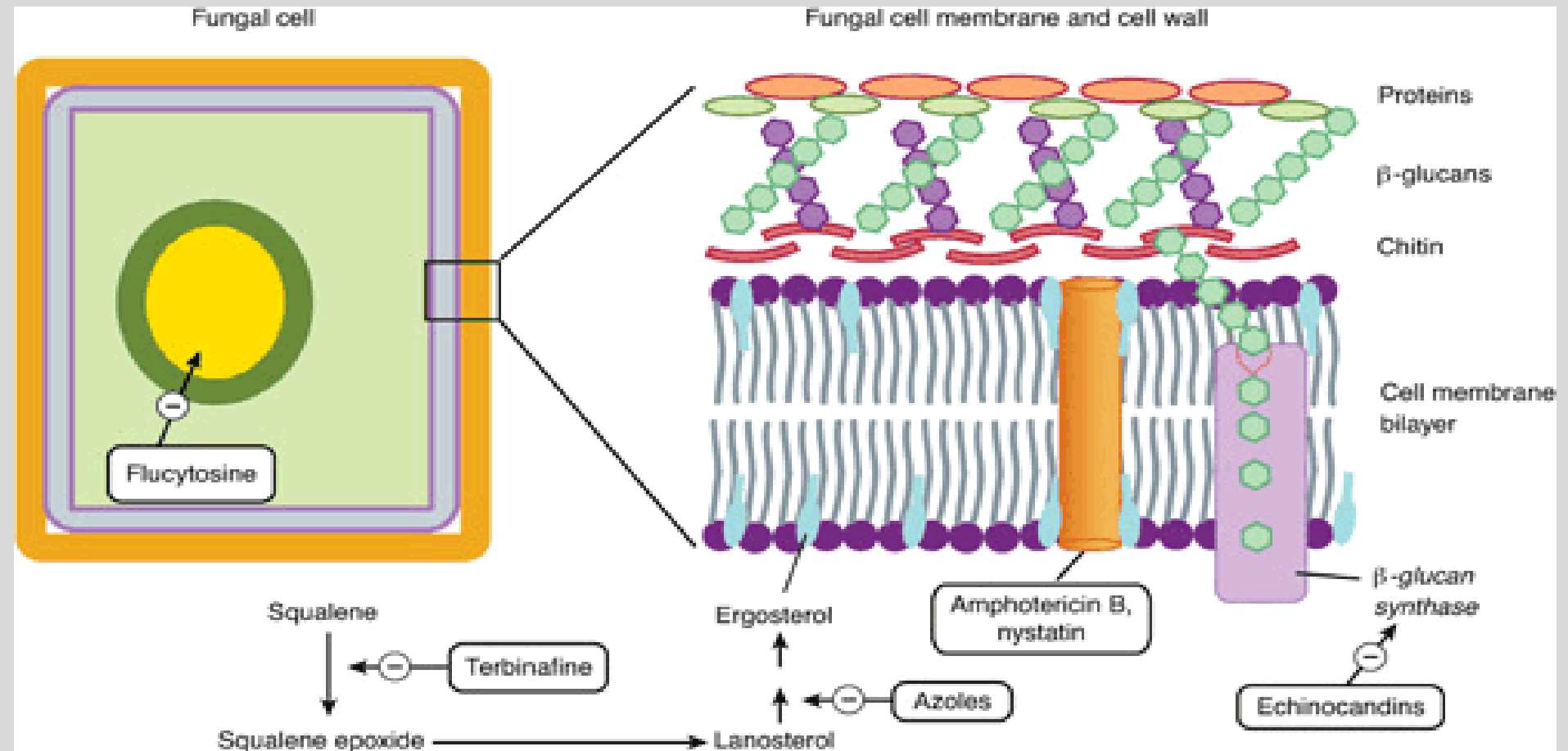
Micafungin

Anidulafungin

Rezafungin



large cyclic peptides linked  
to long-chain fatty acid



**Echinocandins**

## HERE IS HOW TO STATE THE PK PROPERTIES:

IV only, highly protein bound, extensive tissue distribution but poor CNS penetration, slow hepatic non-CYP metabolism, excreted in feces with very little renal clearance → renal dosing adjustment not needed, drug-specific half-life

### Echinocandins

|                        |                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b>               | I.V. only                                                                                                                                                                                                                                                 |
| <b>D</b>               | plasma albumin binding (>97%)<br>extensive tissue distribution but poor distribution to CSF                                                                                                                                                               |
| <b>M</b>               | extensive non-CYP metabolism and spontaneous degradation to amino acid components                                                                                                                                                                         |
| <b>E</b>               | urine / feces primarily as metabolites<br>Dose adjustment for renal impairment not required.                                                                                                                                                              |
| <b>t<sub>1/2</sub></b> | <div>Caspofungin.... 9-11 h, non-linear elimination kinetics</div> <div>Micafungin..... 11-15 h, linear elimination</div> <div>Anidulafungin.. 24-48 h, linear elimination</div> <div>Rezafungin..... ave ~150 h (once weekly dosing), linear elim.</div> |

Echinocandins are noncompetitive inhibitors of  $\beta$ -1,3-glucan synthase. Weakens cell wall  $\rightarrow$  osmotic influx of water  $\rightarrow$  cells swells  $\rightarrow$  rupture

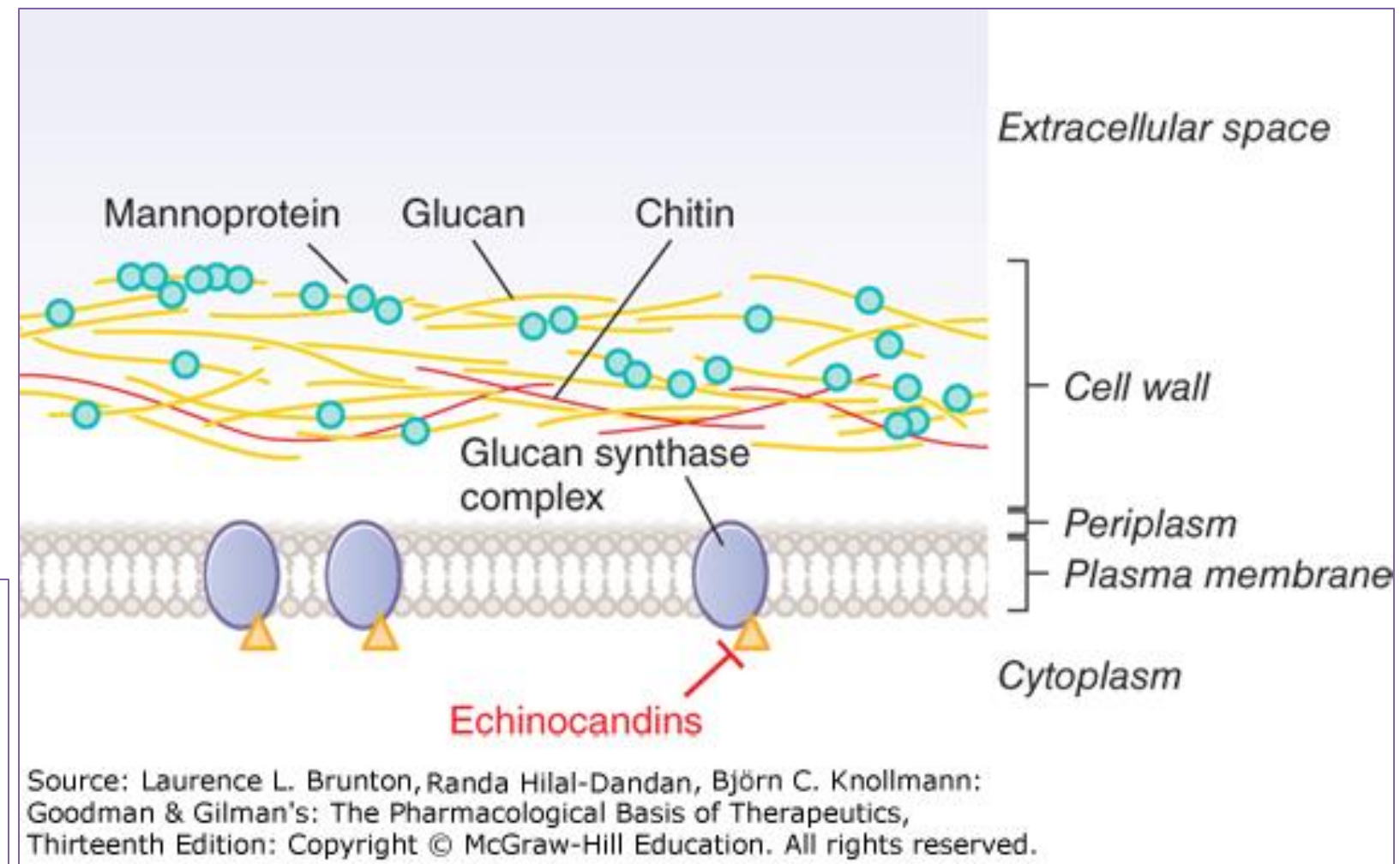
This action disrupts the rigidity, structural integrity, and function of the fungal cell wall, causing cell death.

### **Activity:**

- Narrow spectrum
- *Candida* (fungicidal)  
active against azole-resistant *Candida* spp.
- *Aspergillus* (fungistatic)

### **Resistance (*Candida*): rare to date**

Mutation in *FKS1/2* gene encoding a protein in the glucan synthase complex  $\rightarrow$   $\downarrow$  affinity for the drugs.



The fungal cell wall and membrane and the action of echinocandins. The strength of the fungal cell wall is maintained by fibrillar polysaccharides, largely  $\beta$ -1,3-glucan and chitin, which bind covalently to each other and to proteins. A glucan synthase complex in the plasma membrane catalyzes the synthesis of  $\beta$ -1,3-glucan; the glucan is extruded into the periplasm and incorporated into the cell wall. Echinocandins inhibit the activity of the glucan synthase complex, resulting in loss of the structural integrity of the cell wall. The Fks1p subunit of glucan synthase appears to be the target of echinocandins, and mutations in Fks1p cause resistance to echinocandins.

## Caspofungin Therapeutic Uses: **Serious Infections**

| Drug                           | Candidemia<br>(fungicidal) | Invasive<br>Aspergillosis<br>(fungistatic) | Neutropenic<br>fever | Prophylaxis* |
|--------------------------------|----------------------------|--------------------------------------------|----------------------|--------------|
| Caspofungin<br>adults / peds   | ✓                          | ✓                                          | ✓                    | ✓            |
| Micafungin<br>adults / peds    | ✓                          | ✓                                          | ✓                    | ✓            |
| Anidulafungin<br>adults / peds | ✓                          | ✓                                          | ✓                    | ✓            |
| Rezafungin<br>adults           | ✓                          | effective but<br>not approved              |                      |              |

\*Prophylaxis against invasive fungal infections

Patients at risk for invasive fungal infections include acute leukemia (neutropenia and intensive induction therapy) and hematopoietic stem cell transplantation.

### Adverse Effects

Generally well tolerated

diarrhea, nausea, abdominal pain

- Phlebitis at infusion site
- Histamine-mediated effects: flushing, urticaria, hypotension
- Hepatic impairment: ↑LFTs; hepatitis reported.
- Cardiotoxicity – low rates reported

**Pregnancy:** May cause fetal harm, based on animal reproduction studies: weigh risk/benefit

### Drug interactions: Monitor therapy

- Caspofungin + Cyclosporine → may ↑ each other's levels → may ↑ toxicity of each
- Rifampin and other inducers of clearance may increase caspofungin levels.
- Micafungin + Sirolimus → ↑sirolimus levels
- All of them may decrease therapeutic effects of *Saccharomyces boulardii* probiotic yeast supplement for tx of antibiotic-associated *C. difficile* diarrhea.

## Check your knowledge

What is the fungal cell target of the echinocandins?

What is the spectrum of action of echinocandins?

What is the clinical importance of the echinocandins?

What is the effect of drug interactions with concurrently administered drugs?

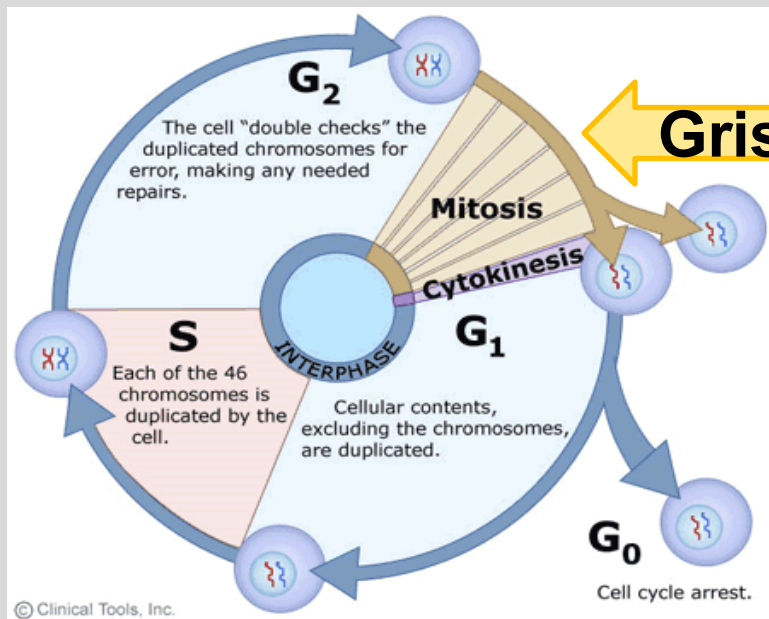
What are the recommendations for use of glucan synthase inhibitors in pregnancy?



# Systemic Antifungal Drugs for Mucocutaneous Infections

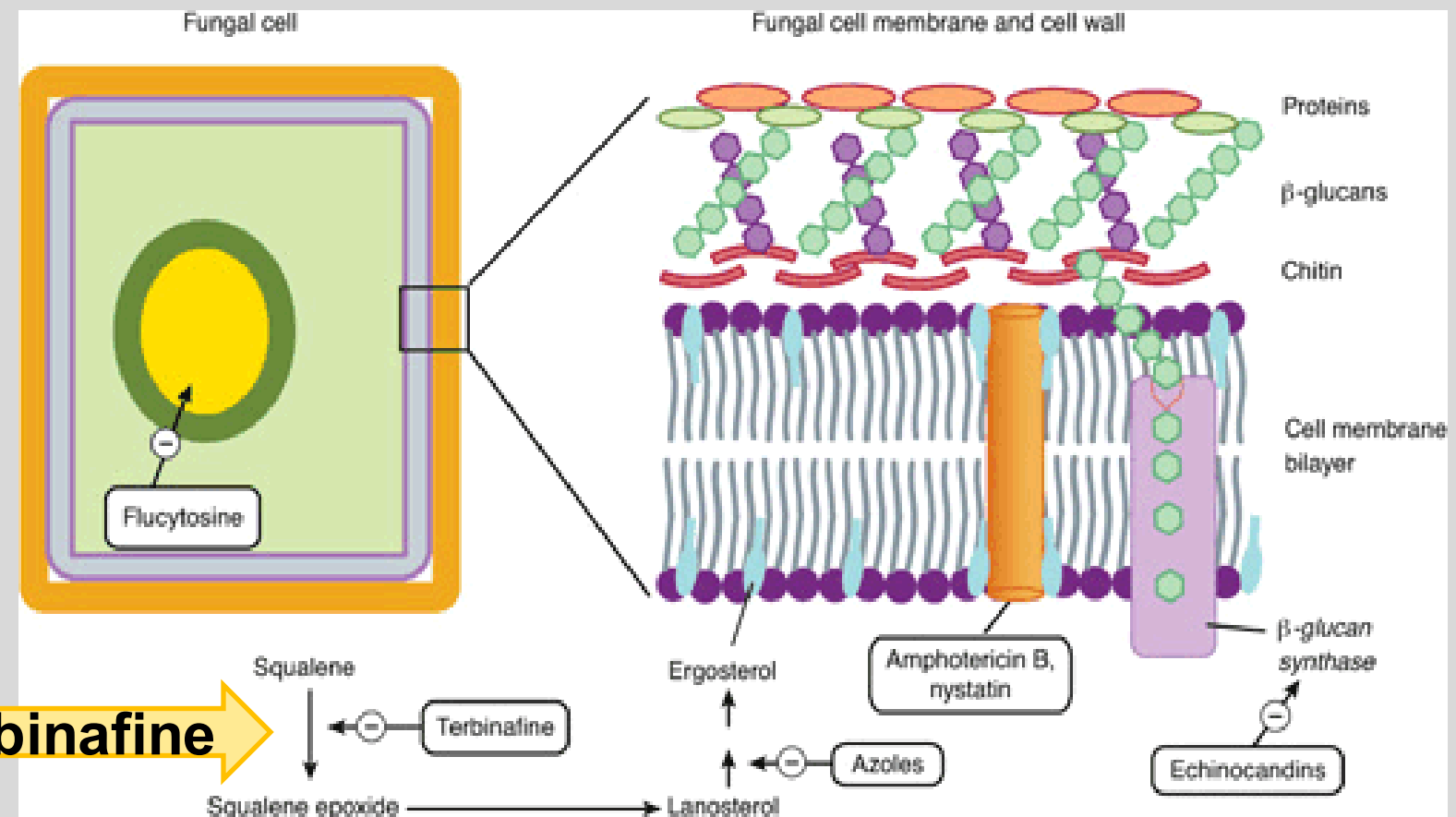
\*Griseofulvin

\*Terbinafine



**Griseofulvin**

**Terbinafine**



## Griseofulvin

- poor oral absorption → low bioavailability
- Absorption is improved with microsize and ultramicrosize formulations and fatty food.
- ***deposits in keratin precursor cells of skin, hair, nails***
  - concentrates in liver, fat, skeletal muscles
  - extensive hepatic metabolism
  - urine (metabolites) and feces
  - $t_{1/2}$  9-24 hours

## Terbinafine

- well absorbed from GI tract
  - 40% bioavailability (marked hepatic first-pass effect)
  - >99% plasma protein bound
  - accumulates in skin, nails, and fat
  - Hepatic metabolism via several CYPs
  - $t_{1/2}$  initial ~12 hours
  - $t_{1/2}$  steady state 200-400 hrs
- ⇒ *not recommended in patients with marked renal or hepatic impairment*

# Griseofulvin accumulates in keratin —Fungistatic—

Binds keratin in keratin precursor cells

Drug-keratin complex is taken up by dermatophyte as hair/skin/nails are replaced

Griseofulvin binds to fungal tubulin

Interferes with microtubule function

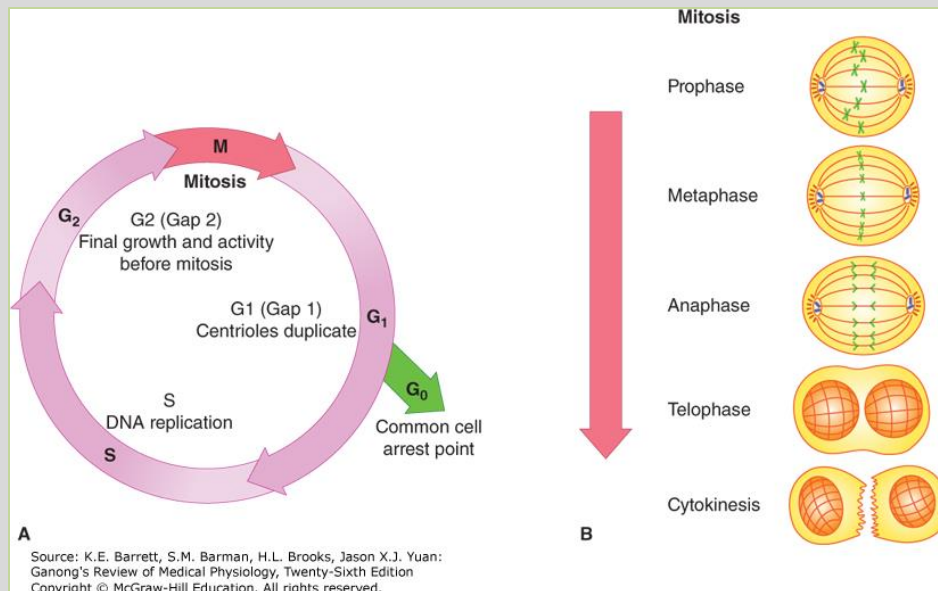
Inhibits fungal cell mitosis at metaphase

## Griseofulvin Treatment

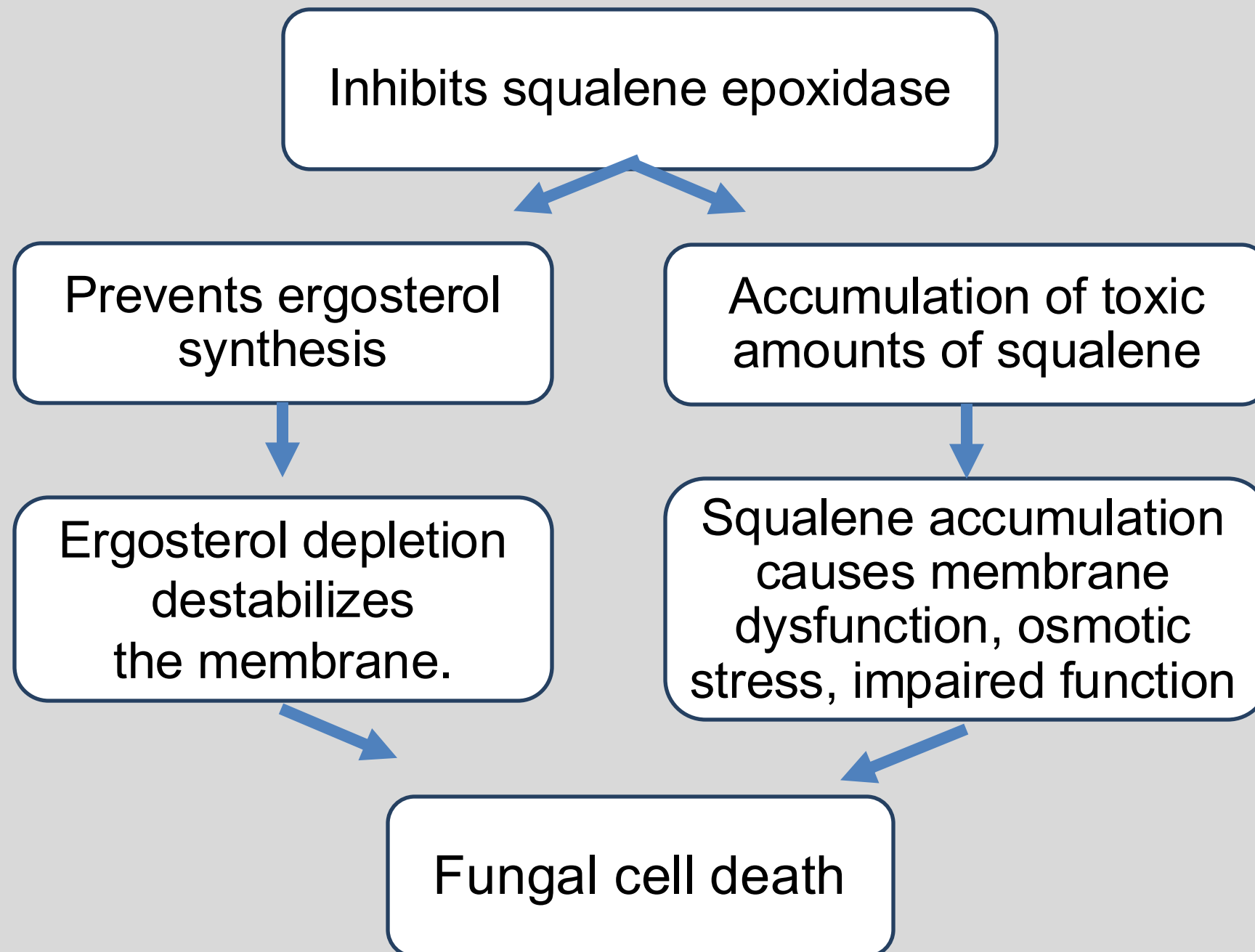
Dermatophytoses (tinea) of skin/scalp, hair, and nails

Treatment duration 6-18 months

Terbinafine has replaced griseofulvin for the treatment of onychomycosis.



## Terbinafine accumulates in keratin —Fungicidal—



### Terbinafine Treatment

Spectrum: Tinea and *Candida*

Drug of choice for

- onychomycosis (oral)
- (more effective than griseofulvin and itraconazole for fungal infections of fingernails and toenails)

Treatment x3 months

- dermatophytoses (topical) and
- cutaneous candidiasis (topical)

- Griseofulvin** Low incidence of serious adverse effects
- Headache, GI upset
  - Rash, photosensitivity
  - Hypersensitivity reactions: rashes, serum sickness, hepatitis, hematologic effects, lupus-like syndrome

**DDIs: CYP inducer**

- ↓ Contraceptive levels → contraceptive failure
- ↓ Warfarin levels

**Contraindications:**

- Pregnancy (teratogenic)
- Hepatic failure
- Porphyria

**Terbinafine** Generally well tolerated

- GI effects, headache, rash
- Taste / smell disturbances

Rare but serious:

- Hepatotoxicity
- Neutropenia
- Hypersensitivity reactions  
Rash, SJS/ TEN

**Drug interactions:**

- minor substrate of several CYPs
- strong inhibitor of CYP2D6

**Contraindications:**

- Chronic or active liver disease
- Pregnancy (limited information)



Review: Compare and contrast griseofulvin and terbinafine.

|                        |
|------------------------|
| ADME                   |
| Mechanism of<br>action |
| Therapeutic uses       |
| Adverse effects        |
| Drug interactions      |
| Contraindications      |

# Topical Antifungal Therapy

There are numerous topical antifungal products. For your reference, they are listed on the following slides.

- Creams, ointments, solutions, powders, vaginal suppositories

Topical antifungal are for the treatment of limited superficial infections of skin and mucous membranes:

- Candidiasis, dermatophytoses (tinea), fungal keratitis
- Penetration into hyperkeratotic lesions is often poor.
- Numerous topical antifungals are available.

The products to know by name for this lecture are:

| Azoles         | Non-Azoles    |
|----------------|---------------|
| * Ketoconazole | * Nystatin    |
| * Clotrimazole | * Terbinafine |

| Drug                                                               | Mechanism of action                                                                                                                               | Spectrum of activity                                                                                      |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| *Nystatin – Polyene                                                | Binds ergosterol forming pores in the plasma membrane                                                                                             | <i>Candida</i> only                                                                                       |
| *Azoles                                                            | 14- $\alpha$ -demethylase inhibitors                                                                                                              | Fungistatic<br><i>C. albicans</i> , Dermatophytes,<br><i>Malassezia furfur</i><br>(pityriasis versicolor) |
| Allylamines:<br>*Terbinafine, Naftafine<br>Benzylamine: Butenafine | Squalene epoxidase inhibitors                                                                                                                     | Fungicidal<br>Dermatophytes<br><i>Malassezia furfur</i>                                                   |
| Ciclopirox                                                         | Chelates trivalent metal ions → inhibits metal-dependent enzymes; also blocks intracellular transport, which disrupts DNA, RNA, protein synthesis | Fungicidal<br><i>C. albicans</i> , Dermatophytes,<br><i>Malassezia furfur</i>                             |
| Tolnaftate                                                         | Distorts the hyphae and stunts growth of susceptible fungi; may involve inhibition of squalene epoxidase                                          | Fungicidal<br>Dermatophytes<br><i>Malassezia furfur</i>                                                   |
| Tavaborole                                                         | Inhibits leucyl-tRNA synthase → protein synthesis inhibitor                                                                                       | Fungistatic<br>Trichophyton                                                                               |

# Therapeutic Uses of Topical Antifungal Agents

| Infection                                                | Azole Treatments                                                                                                                                                                                                                                                                                             | Non-Azole Treatments                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oral candidiasis                                         | <ul style="list-style-type: none"> <li>*Clotrimazole troches (lozenge)</li> <li>Miconazole buccal tablets                             <ul style="list-style-type: none"> <li>Dissolve in mouth (not chewed or swallowed)</li> <li>Treatment and prophylaxis</li> <li>Pleasant-tasting</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>*Nystatin oral suspension                             <ul style="list-style-type: none"> <li>Swish and swallow, or swish and spit</li> <li>Paint mucous membranes of infant's mouth</li> <li>Unpleasant taste even though it contains sucrose</li> </ul> </li> </ul> |
| Vulvovaginal candidiasis, uncomplicated                  | Vaginal creams and suppositories                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>*Nystatin vaginal tablets (must be prepared by compounding pharmacy)</li> </ul>                                                                                                                                                                                      |
|                                                          | <ul style="list-style-type: none"> <li>*Clotrimazole      Terconazole</li> <li>Miconazole        Tioconazole</li> <li>Butoconazole</li> </ul>                                                                                                                                                                |                                                                                                                                                                                                                                                                                                             |
| Cutaneous candidiasis                                    | Various products for diaper dermatitis and intertrigo                                                                                                                                                                                                                                                        | *Nystatin cream for diaper dermatitis and intertrigo                                                                                                                                                                                                                                                        |
| Dermatophyte and <i>Malassezia</i> infections, cutaneous | Various azole products for Trichophyton, Epidermophyton<br>Tinea pedis / cruris / corporis / versicolor                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>*Terbinafine      Ciclopirox</li> <li>Naftafine        Tolnaftate</li> <li>Butenafine</li> </ul>                                                                                                                                                                     |
| Onychomycosis                                            | Efinaconazole solution                                                                                                                                                                                                                                                                                       | Ciclopirox lacquer<br>Tavaborole solution                                                                                                                                                                                                                                                                   |
| Seborrheic dermatitis: <i>Malassezia</i>                 | Scalp: *Ketoconazole shampoo<br>Non-scalp: various topical creams, gels, or foam                                                                                                                                                                                                                             | Ciclopirox<br>Zinc pyrithione<br>Selenium sulfide                                                                                                                                                                                                                                                           |

## Mild oral candidiasis treatments



Nystatin oral  
suspension  
(swish and spit)



Clotrimazole troche  
(dissolve in mouth)

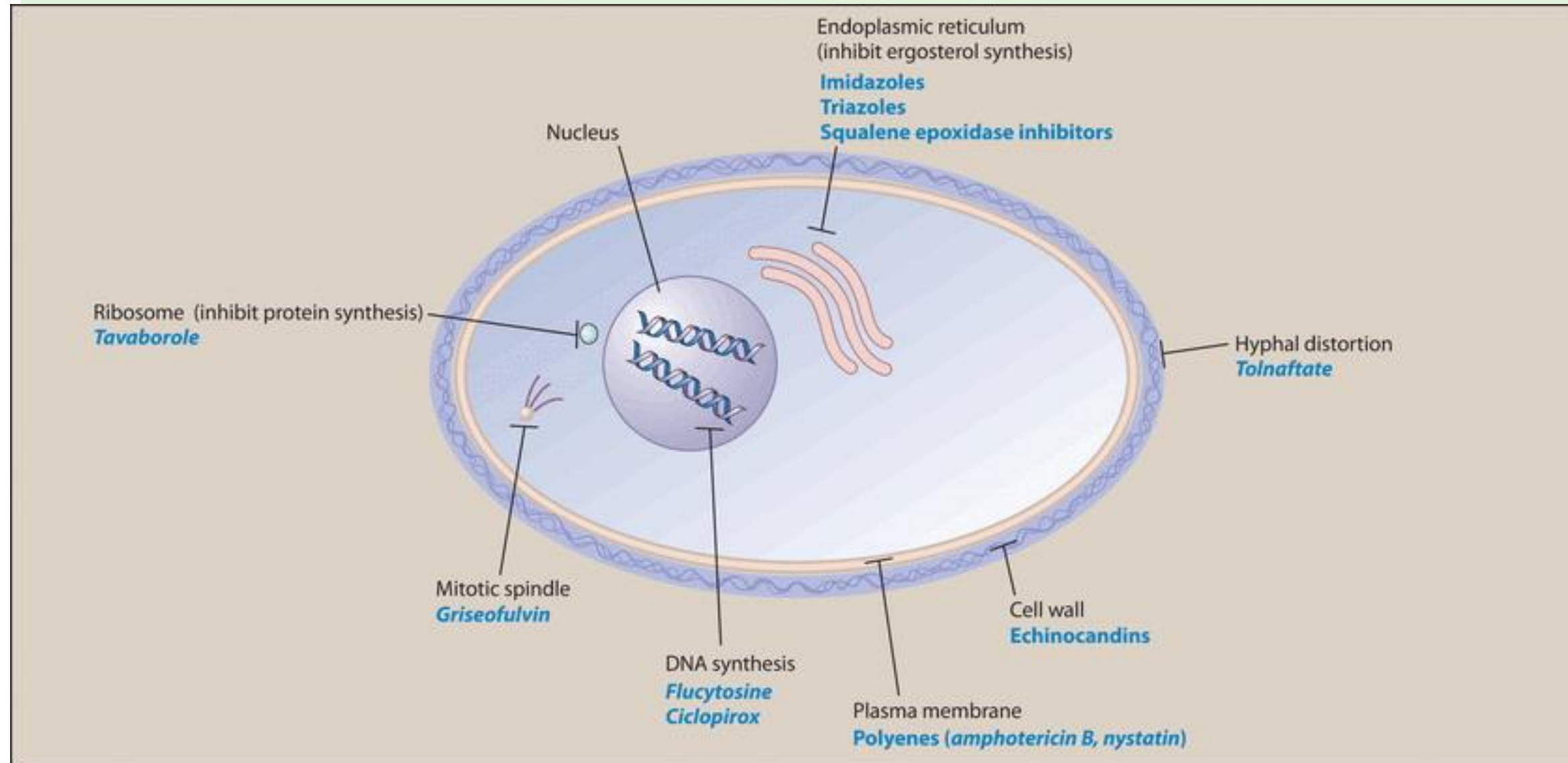




*Candida albicans*—thrush in mouth. Note whitish plaques on tongue.

# Summary of Antifungal Agents

From: **33 Antifungal Drugs, Figure 33.3**  
Lippincott® Illustrated Reviews: Pharmacology, 8e, 2022



Cellular targets of antifungal drugs.

Mycotic infections may involve the skin or subcutaneous tissues or may cause systemic infections.

Fungi are eukaryotic with rigid cell walls composed of chitin. Their membrane contains ergosterol, rather than cholesterol. These structures are useful targets of drug therapy.

Amphotericin B (AMB):

- Broad spectrum of activity; fungicidal
- AMB is a polyene antibiotic binds ergosterol in fungal cell membrane causing pore formation and leakage of cell components.
- Relative selectivity: Ergosterol has greater affinity for AMB than cholesterol has.
- Resistance is rare; possibly due to mutations in ERG gene or replacement of ergosterol with precursor sterols.
- AMB is used for the treatment of serious systemic and CNS fungal infections, including during pregnancy.
- Nephrotoxicity and infusion reactions (fever, chills, tachypnea) limit its use.

Flucytosine (5-FC)

- Prodrug converted to cytotoxic antimetabolite, 5-fluorouracil (5-FU), which is activated by phosphorylation and inhibits fungal RNA and DNA synthesis.
- Narrow spectrum of activity; fungistatic
- Resistance develops rapidly when it is used as a single agent, probably by reduced cytosine permease or cytosine deaminase activity.
- 5-FC is an adjunct in the treatment of cryptococcal meningitis, and some invasive candidiasis
- Generally well tolerated. Concentration-dependent leukopenia, nausea, vomiting, diarrhea. Monitor renal function.

Azole antifungal agents: imidazoles and triazoles.

- Broad spectrum of activity, in general; fungistatic
- PK: Most are substrates and strong inhibitors of CYP3A4 and other CYPs.
- Mechanism: Competitive inhibitors of 14- $\alpha$ -demethylase (CYP51), blocking the conversion of lanosterol to ergosterol leading to membrane instability and inhibits fungal cell growth.
- Resistance: *ERG11* gene mutation, leading to decreased binding affinity of 14- $\alpha$ -demethylase and decreased azole efficacy. Some fungal strains upregulate efflux pumps or increase production of lanosterol.
- Triazoles are generally used for treatment of systemic and cutaneous mycoses, while imidazoles are topical agents for cutaneous infections.
- QT interval prolongation is a potentially serious adverse effect of all azole antifungals. Other side effects include GI disturbances and rash. There is a high risk of drug interactions.

Echinocandins

- Narrow spectrum; *Candida* spp (fungicidal), *Aspergillus* spp (fungistatic)
- Mechanism: Noncompetitive inhibition of  $\beta$ -1,3-glucan synthase, which decreases cell wall synthesis.
- Resistance occurs by FKS-1 and FKS-2 alterations that result in decreased affinity for the echinocandins.
- The drugs are well tolerated. Flushing may occur. Caspofungin increases systemic concentrations of the immunosuppressants cyclosporine and tacrolimus, and micafungin increases sirolimus levels. Cautious use of the echinocandins is recommended in patients taking these immunosuppressants.



## Terbinafine

- Broad spectrum, fungicidal
- Mechanism: Noncompetitive inhibition of squalene epoxidase, which prevent ergosterol biosynthesis, leading to cell death.
- Resistance is reported and linked to mutations in squalene epoxidase gene, SQLE
- Terbinafine accumulates in keratin-rich tissues and is more effective than griseofulvin in the treatment of dermatophytoses of the nails, skin, and scalp; available for oral and topical use.
- Side effects include GI upset and headache. Terbinafine it is contraindicated in patients with chronic or active hepatic disease. Terbinafine is a CYP substrate and CYP2D6 inhibitor.

## Griseofulvin

- Narrow spectrum – dermatophytes; fungistatic
- Binds human keratin in skin, hair, and nails and causes disruption of the mitotic spindle and inhibition of mitosis
- Treatment of dermytophytoses of the scalp and hair. It has been largely replaced by terbinafine for the treatment of onychomycosis. Long duration of therapy – 6-18 months.
- Ultramicrosize griseofulvin is adequately absorbed and absorption is enhanced by fatty food.
- Side effects include GI upset, headache, rash, and photosensitivity, It is contraindicated in pregnancy and patients with porphyria or hepatic failure.

## Topical antifungal agents

- A number of antifungal drugs are used topically for superficial infections caused by *C. albicans* and dermatophytes.
- Preparations include oral suspension, oral troches, vaginal creams and suppositories, skin creams, ointments, and powder.

### Nystatin

- Polyene antibiotic that disrupts fungal membranes by binding to ergosterol.
- (Recall: Amphotericin B is a polyene antibiotic.)
- Topically to suppress local *Candida* infections, including mild oral-pharyngeal candidiasis (thrush)
- Orally for secondary prevention of fungal peritonitis in patients on peritoneal dialysis
- Nystatin is poorly absorbed from the gastrointestinal tract.
- Nephrotoxic (like AMB), therefore nystatin is not administered systemically.

### Azole compounds:

- miconazole, clotrimazole, ketoconazole, and several others and terbinafine topical products are used for superficial tinea infections and mucocutaneous *Candida* infections.

Terbinafine, naftifine, butenafine: Squalene epoxidase inhibitors topically for tinea infection and cutaneous candidiasis

Tolnaftate: Distorts hyphae and stunts fungal growth for treatment of tinea infections

Ciclopirox: Inhibits transport of essential molecules and interferes with enzyme functions, DNA repair, and cell division signals and structures.

Tavaborole: tRNA synthase inhibitor for treatment of onychomycosis.

# Fungal Infection



## Check your knowledge

What are the advantages of liposomal AMB and AMB lipid complex compared to AMB deoxycholate?

- ***Infusion-related toxicity and nephrotoxicity are reduced***, which permits use of larger doses compared to conventional AMB.
- AMB lipid formulations enhance delivery to the site of infection while reducing organ toxicity.
- Higher doses can improve effectiveness, particularly in severe or resistant fungal infections.

What is AMB's mechanism of antifungal action?

- AMB associates with ergosterol in fungal membrane causing pores to form. Cations and other small molecules leak out.
- Ergosterol is sequestered further compromising membrane integrity, leading to fungal cell death.

What are the toxicities associated with AMB administration?

- Infusion reactions, nephrotoxicity (azotemia, acute kidney injury, renal tubular acidosis), and anemia (impaired EPO synthesis)

## Check your knowledge

What physical properties of the amphotericin B molecule are probably important for its fungus-killing activity?

- The amphipathic property allows the drug to interact with both hydrophilic polar head groups of the cell membrane and hydrophobic membrane core and ergosterol embedded in the membrane.
- The amphoteric property (acid and base functional groups) enhances binding to ergosterol and helps to sequester ergosterol, further destabilizing the cell membrane.

What accounts for its relatively selective toxicity against fungi?

- AMB affinity for ergosterol in fungal membranes is many times greater than its affinity for cholesterol in mammalian membranes.
- What are the likely mechanisms of resistance?
- Reduced concentrations of ergosterol due to mutations in various ERG genes or replacement with sterol precursors. Resistance to AMB is rare.

AMB is an important therapeutic for what patients and infections?

- AMB is primarily for treatment of patients with ***progressive and potentially life-threatening fungal infections***.
- AMB should ***not*** be used to treat noninvasive forms of fungal disease such as oral thrush, vaginal candidiasis, and esophageal candidiasis in patients with normal neutrophil counts.



## Check your knowledge

### What is the basis of flucytosine's selectivity?

- Mammalian cells cannot activate 5-FC as they do not have cytosine permease or cytosine deaminase.

### What is the basis of flucytosine's toxicity?

- Intracellular conversion to toxic 5-FU in fungal cell

### How does flucytosine induce cellular cytotoxicity?

- 5-FUTP incorporation of FUTP into RNA → ↓RNA synthesis (thus, ↓protein synthesis)
- 5-FdUMP inhibits thymidylate synthase → ↓ DNA synthesis

### What are the therapeutic uses of 5-FC and why?

- Narrow spectrum of activity
- Cryptococcal meningitis with AMB and selected invasive candidiasis with AMB.
- Oral, crosses BBB; rapid development of resistance when used alone

### What are major adverse effects of 5-FC and how can they be reduced?

- Well tolerated when renal function is normal – Adverse effects are concentration-dependent
- Leukopenia, thrombocytopenia / elevated liver function enzymes (AST/ALT) / nausea, vomiting diarrhea
- Monitor renal function

## Check your knowledge

Which drug is first line for treatment of fungal infections by dimorphic fungi?

- Itraconazole

Which drug is first line for consolidation and maintenance of cryptococcal meningitis?

- Fluconazole
- Alternatives are voriconazole, posaconazole and isavuconazole (isavuconozium is the prodrug).
- Initial therapy regimen is amphotericin B + flucytosine

Which drug is the preferred initial treatment for invasive aspergillosis?

- Voriconazole

Which drug is an alternative for the initial treatment of invasive aspergillosis in a patient with optic neuritis or peripheral neuropathy?

- Posaconazole or isavuconazole and they are better tolerated than voriconazole.
- Voriconazole can cause ocular, neurological, and severe dermatologic toxicities.

Which drugs are used as step-down therapy for mucormycosis?

- Posaconazole and isavuconazole
- Amphotericin B is the preferred initial therapy.

Which oral drug is first line for vulvovaginal candidiasis?

- Fluconazole

## Check your knowledge

### What adverse effects can occur with all of the azole antifungals?

- Minor GI disturbances (eg nausea, diarrhea); ↑ serum aminotransferases (hepatic enzymes), hepatitis (rare); prolongation of QT interval; TdP/ventricular arrhythmias; metabolic-related hypokalemia, pseudoaldosteronisms and hypertension; immediate and delayed hypersensitivity reactions

### Which azole antifungal should be avoided in patients with a history of heart failure?

- Itraconazole has negative inotropic effects (reduced force of contraction) and can worsen heart failure AND it is associated with new heart failure and with new and worsening hypertension. Metabolic-related effects observed more often than with other triazoles – except posaconazole.

### Which azole antifungal is associated with optic neuritis and other ocular toxicities?

- Voriconazole: Optic neuritis, scleritis, episcleritis, photophobia

### Which azole antifungal is associated with peripheral neuropathy?

- Voriconazole: Numbness and tingling in the extremities may be rapidly debilitating and irreversible. Encephalopathy is reported.

### Which azole antifungal is most likely to interfere with steroidogenesis?

- Ketoconazole, an imidazole antifungal that is currently used only topically, has a high risk of adrenal suppression due to its strong inhibition of steroidogenesis.
- Itraconazole and posaconazole have a moderate risk of adrenal suppression.
- FYI: Cortisol, aldosterone, and the androgen precursors of sex hormones, dehydroepiandrosterone (DHEA) and DHEA sulfate are synthesized in the adrenal cortex.

## Check your knowledge

What is the major mechanism of drug interactions with the azole antifungal drugs?

- CYP inhibition. All azole antifungals are strong CYP3A4 inhibitors, except fluconazole and isavuconazole, which both inhibit CYP3A4 but with moderate affinity for the enzyme.

One of the azole antifungals is cleared renally and excreted unchanged in urine. Which one? What do we need to think about in terms of dosing this drug?

- Fluconazole, 80% of the dose is excreted unchanged in the urine.
- Dosage adjustment for patients with renal insufficiency.
- Note: Fluconazole is a strong inhibitor of CYP2C19 and inhibit CYP2C9 and 3A4 with lower affinity.

Two triazole antifungal drugs are formulated with cyclodextrin to solubilize the hydrophobic drugs for intravenous administration. What toxicity can occur in patients with renal insufficiency? What are the two drugs?

- Accumulation of cyclodextrin in patients with renal insufficiency can cause toxicity to the kidney tubular cells. The drugs are voriconazole and posaconazole.

One of the antifungals in this group does not require dosage adjustment for patients with renal insufficiency or liver impairment. What is it? Why are dosage adjustments not required?

- The drug is posaconazole because it is minimally metabolized (~17% by UGT) by the liver, excreted unchanged in the feces, and is not cleared through the kidney.

## Check your knowledge

### What is the fungal cell target of the echinocandins?

- They inhibit  $\beta$ -1,3-glucan synthase complex, which disrupts the integrity of the fungal cell wall, leading to cell death.

### What is the spectrum of action of echinocandins?

- *Candida* spp (fungicidal); *Aspergillus* spp (fungistatic)

### What is the clinical importance of the echinocandins?

- Treatment and prophylaxis of serious *Candida* infections; treatment of invasive aspergillosis not responsive to other drugs

### What is the effect of drug interactions with concurrently administered drugs?

- Caspofungin and cyclosporine may increase plasma concentrations of each other increasing the risk of hepatotoxicity (echinocandins) and nephrotoxicity (cyclosporine)
- Micafungin increases sirolimus levels increasing the risk of numerous toxicities.
- All may diminish the therapeutic effects of *Sacharomyces boulardii*, a beneficial yeast in the gut microbiome.
- Ibrexafungerp: CYP3A4 inducers (avoid) and inhibitors (caution)

### What are the recommendations for use of glucan synthase inhibitors in pregnancy?

- Echinocandins: Animal reproduction studies suggest the potential for fetal harm. Use only if benefit to the mother outweighs the risk to the fetus.



## Review: Compare and contrast griseofulvin and terbinafine.

|                     |                                                                                                                            |                                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| ADME                | Oral, once daily<br>G: Take with fatty food                                                                                | Extensive hepatic metabolism<br>Terminal $t_{1/2}$ :<br>G: 9-24 h; T: 200-400 h at steady state |
| Mechanism of action | G: binds microtubules in new keratin and prevents mitosis<br>T: squalene epoxidase inhibitor reducing ergosterol synthesis |                                                                                                 |
| Therapeutic uses    | Dermatophyte infections<br>G: DOC for tinea capitis in children<br>T: DOC for onychomycosis<br>DOC: drug of choice         |                                                                                                 |
| Adverse effects     | Both generally well tolerated<br>GI disturbances, hypersensitivity reactions                                               |                                                                                                 |
| Drug interactions   | G: CYP inducer<br>T: Strong CYP2D6 inhibitor                                                                               |                                                                                                 |
| Contraindications   | G: liver failure, porphyria, pregnancy<br>T: chronic or acute liver disease, avoid in pregnancy                            |                                                                                                 |

- Access Medicine Ganong's Review of Medical Physiology, 26e, 2019; General Principles
- Access Medicine Goodman and Gilman's The Pharmacological Basis of Therapeutics 14e, 2023; Chapter 61: Antifungal Agents
- Access Medicine Harrison's Principles of Internal Medicine, 21e, 2022; Chapter 211: Pathogenesis, Diagnosis, and Treatment of Fungal Infections
- Access Medicine Jawetz, Melnick, & Adelberg Medical Microbiology, 28e, 2019; Chapter 45: Medical Mycology
- Access Medicine Katzung's Basic & Clinical Pharmacology 16e, 2024; Chapter 48: Antifungal Agents
- Clinical Key Murray, Rosenthal, and Pfaller *Medical Microbiology* 9e, 2022; Chapter 61: Antifungal Agents
- UpToDate Lexidrug monographs

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